

Brownian loop measure on hyperbolic pair of pants

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Preface

In Section 1, we introduce the notion of the Brownian loop measure and present the main question of this thesis together with its answer. Section 2 provides a brief review of hyperbolic geometry; readers already familiar with the subject may skip this chapter. In Section 3, we describe the construction and pasting of hyperbolic surfaces and include a subsection on perpendiculars. This theory plays a crucial role in establishing the existence of hyperbolic pairs of pants with prescribed closed boundary geodesic lengths in Section 5.2.

In Section 4, we study the orbit paths of hyperbolic, parabolic, and elliptic elements of $PSL(2, \mathbb{R})$ and their actions on the hyperbolic half-plane $\mathbb{H}$. Section 5 is devoted to constructing a hyperbolic pair of pants for any given $(a, b, c) \in \mathbb{R}_+^3$ (the lengths of the closed geodesics), using right-angled hyperbolic hexagons.

In Section 6, we introduce the Uniformization theorem and establish the one-to-one correspondence between essential free homotopy classes of closed curves in a hyperbolic surface with totally geodesic boundary and conjugacy classes of hyperbolic elements in the associated Fuchsian group.

In Section 7, we extend the notion of the Brownian loop measure to hyperbolic surfaces. Section 8 and Section 9 are devoted to attempts at resolving the main question introduced in Section 1. Finally, in Section 10, we present an alternative and more effective approach to computing the Brownian loop measure of a hyperbolic funnel.

The appendix collects the topological tools used throughout this work. The thesis is designed to be largely self-contained. The initial chapters are based on cited resources. All mistakes are completely mine.

To my grandparents,

Dadu & Dida.

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1 Introduction

Randomness and geometry have long been intertwined in the study of two-dimensional models of statistical mechanics. The discovery that scaling limits of the interfaces in the two dimensional models in statistical mechanics often exhibit conformal invariance has been a central theme of research in the recent decades. A breakthrough in this direction was done by Oded Schramm [6] when he introduced the Schramm-Loewner Evolution (SLE) which provides a natural candidate for conformally invariant scaling limits of random interfaces. Smirnov's celebrated result that the scaling limit of the exploration process in the honeycomb lattice is SLE_6 [8] displayed the power of this theory. However, the theory of SLE still doesn't capture the full scaling limit.

Here, comes the role of Brownian loop soup, introduced by Lawler and Werner [4], which has emerged as a fundamental object to capture the full scaling limit. The Brownian loop soup not only provides a universal way to generate conformally invariant collections of loops appearing in the full scaling limit but is also indispensable for understanding objects ranging from loop-erased random walks to the geometry of Liouville quantum gravity. For example, Werner has shown how the Brownian loop soup can generate a whole family of SLE loops [10], while Camia and Newman [2] constructed the full scaling limit of critical percolation using loop soups. At the heart of the Brownian loop soup is the concept of Brownian loop measure.

Concretely, a rooted Brownian loop in $\mathbb{C}$ is a continuous function $\tilde{\gamma} : [0, t_{\tilde{\gamma}}] \rightarrow \mathbb{C}$ with $\tilde{\gamma}(0) = \tilde{\gamma}(t_{\tilde{\gamma}})$, $t_{\tilde{\gamma}}$ the time duration of the loop and $\tilde{\gamma}(0)$ the root. We do two things with any such loop $\tilde{\gamma}$: we translate the root of $\tilde{\gamma}$ to 0 and we perform a Brownian scaling of it so that the time duration of the loop is now 1. We call this new loop γ i.e.

$$\gamma(s/t_{\tilde{\gamma}}) = (\tilde{\gamma}(s) - z)/\sqrt{t_{\tilde{\gamma}}}, \quad 0 \leq s \leq t_{\tilde{\gamma}}.$$

If we call $\mathcal{C}$ to be the set of all rooted loops in $\mathbb{C}$ and $\mathcal{C}_t$ to be the set of all rooted loops with $t_{\tilde{\gamma}} = t$, $\tilde{\gamma}(0) = \tilde{\gamma}(t_{\tilde{\gamma}}) = 0$, then, the above transformation is a one-to-one correspondence from $\mathcal{C}_1 \times \mathbb{C} \times (0, \infty)$ to $\mathcal{C}$ via the map $(\gamma, z, t) \mapsto \tilde{\gamma}$. Now, introduce the measure

$$\mu = \mu^{br} \times \text{area} \times \left(\frac{1}{2\pi t^2} \right) dt \tag{1}$$

on $\mathcal{C}_1 \times \mathbb{C} \times (0, \infty)$.

Here, μ^{br} is the measure induced on continuous loops from 0 to 0 by a complex Brownian bridge of time duration 1. We realise this as a measure on $\mathcal{C}$ by the one-to-one correspondence. We call this realised measure the rooted 'Brownian loop measure'. Denote the unrooted Brownian loop measure by μ^u . This is the induced measure on equivalence classes of rooted loops where $\eta \sim \xi$ (some rooted loops) if they have the same trace and orientation and different roots. An unrooted Brownian loop soup with intensity λ is a Poissonian realisation from $\lambda\mu^u$.

Our focus

We will study Brownian loop measure in the setting of hyperbolic geometry, particularly in the case of the degenerate *hyperbolic pair of pants*. Consider the *hyperbolic pair of pants* $X = P_{(0,0,c)}$ which is a pair of pants with two cusps and a closed boundary geodesic of length c (see Figure 1). In Section 5.2, we will construct the hyperbolic pair of pants. By construction, it will be locally isometric to $\mathbb{H}$ (hyperbolic upper-half plane model) and hence, have a constant curvature of -1. Call the associated Riemannian metric on X g . Now, remove the boundary and consider X° . This is a Riemann surface (refer Section 6 for details on Riemann surfaces) without boundary and so by a corollary of the *Uniformization theorem* (mentioned in corollary 6.6), $\exists G < \text{Aut}(\mathbb{H})$ such that G acts freely and properly discontinuously on $\mathbb{H}$ and $X^\circ = \mathbb{H}/G$. However, it is not complete with respect to the metric g as we have removed the limit points lying on the boundary.

In Remark 6.13, we will show that there exists a unique Riemannian metric h conformal to g i.e. $h = \rho \cdot g$ where ρ is some positive function $\rho : X^\circ \rightarrow \mathbb{R}_+$, such that the metric h makes X° complete and equips it with a constant curvature of -1. What the metric h does is that it pulls the end of the surface (from where the boundary has been removed) to ∞ (see Figure 2).

Let γ be the closed boundary geodesic of X . Then, let

$$C_X(\gamma) = \{\eta \mid \eta \text{ is a closed curve in } X \text{ freely homotopic to } \gamma\}$$

Let $\tilde{\gamma}$ be the *unique closed geodesic in the essential free homotopy class* $C_X(\gamma)$ in the surface $\tilde{X}$ equipped with the metric h . We denote the length of $\tilde{\gamma}$ computed using h by $l_h(\tilde{\gamma})$.

Main Question: What can we say about $l_h(\tilde{\gamma})$ in terms of $l_g(\gamma)$?

Main result: Using Brownian loop measure of the pair of pants $P_{(0,0,c)}$ glued with a hyperbolic funnel of the same closed boundary length and opposite orientation, we will show in Section 8 that $l_h(\tilde{\gamma}) > l_g(\gamma)$. We refine our result and prove a more stronger result in Section 9 after calculating the Brownian loop measure of the Hyperbolic funnel. There, we show that

$$l_h(\tilde{\gamma}) > \log(1 + e^{l_g(\gamma)} - e^{-l_g(\gamma)}). \quad (2)$$

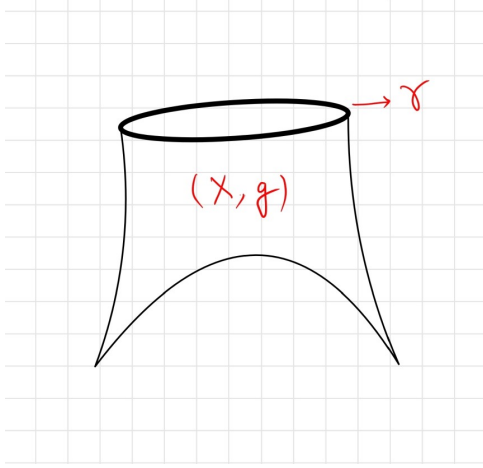


Figure 1: The hyperbolic pair of pants X with closed boundary geodesic γ of length c .

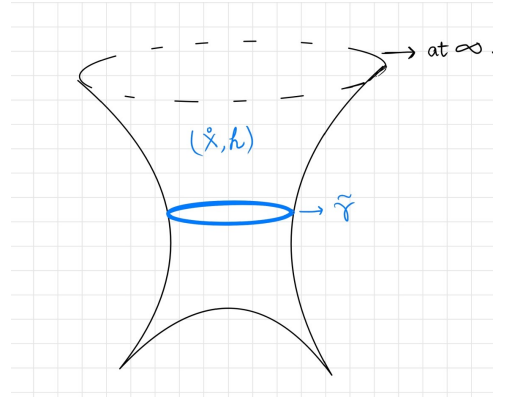


Figure 2: The interior of X equipped with h takes the open end to ∞ and makes it complete.

Motivation for the definition of the rooted Brownian loop measure (Optional).

To motivate the Brownian loop measure in eq. (1), it is useful to begin with its discrete analogue. Consider $\mathbb{Z}^d$ and a loop l here:

$$l = (z_0, \dots, z_n), \quad z_i \sim z_{i+1} \quad \& \quad z_0 = z_n.$$

We denote the length in this case by $|l|$. So, $|l| = n$ here. A natural choice of a measure on unrooted loops is the one that puts a probability of $1/2d$ on each edge independently as at each vertex, there are $2d$ possible directions. Let ν be that measure. Then,

$$\nu(l) = (2d)^{-|l|}.$$

We denote the measure on rooted loops by $\tilde{\nu}$. Apart from seeing an unrooted loop as a rootless loop, seeing it as an equivalence class of rooted loops sharing the same trace and the orientation but different root is mathematically very helpful. This gives us

$$\tilde{\nu}(l) = (2d)^{-|l|}/|l|.$$

We define the *Random walk loop-soup* of intensity λ to be a Poissonian realisation from $\lambda\nu$.

Let A be any measurable set of rooted loops in $\mathcal{C}$. Then, by eq. (1),

$$\mu(A) = \int_{\mathbb{C}} \int_0^\infty \frac{1}{2\pi t^2} \mu_t^{br}(z) 1(A) dt \, dz.$$

Why does this definition make sense?

Justification I

The transition density of the planar Brownian motion in $\mathbb{C}$ starting from any $x \in \mathbb{C}$ wrt the Lebesgue measure is

$$p^{\mathbb{C}}(x, y; t) = \frac{1}{(\sqrt{2\pi t})^2} \exp\left(-\frac{|x - y|^2}{2t}\right).$$

For any rooted loop $\tilde{\gamma}$ with time duration t , $\tilde{\gamma}(0) = \tilde{\gamma}(t)$. So, transition density is $1/(2\pi t)$ in that case. We can think of the Brownian loop measure as the Brownian bridge measure weighted by the transition density and the time duration of the loop i.e.

$$\mu = \mu^{br} \frac{p^{\mathbb{C}}(z, z; t)}{t} dt.$$

Justification II

One can obtain the (unrooted) Brownian loop measure as a weak limit of (unrooted) random walk loop measures with a bit of care as these are infinite measures. However, we recall the idea of a much stronger result here than weak convergence. This result was proved by Lawler and Ferreras.

Consider $(\mathcal{A}_\lambda, \tilde{\mathcal{A}}_\lambda)$ on the same probability space where $\mathcal{A}_\lambda, \tilde{\mathcal{A}}_\lambda$ are realisations of the Brownian loop soup and the Random walk loop soup respectively with the same intensity λ .

Next through Brownian scaling, they scale down the loops of both $\mathcal{A}_\lambda$ and $\tilde{\mathcal{A}}_\lambda$ by a factor of $1/N$ where $N \in \mathbb{N}$ and show that in any bounded domain, after excluding the loops that are ‘too small’, there is a one-to-one correspondence between the Brownian loops and the Random walk loops in the lattice $\frac{1}{N}\mathbb{Z}^2$ and in any such pair, the loops are very close to each other i.e. the supremum norm of the difference is $O(N^{-1} \log N)$ and the difference of their time durations is bounded above by $1/N^2$. This justifies that the Brownian loop measure as defined in eq. (1) is the natural analogue of the Random loop measure. More details about this is present in [3]

Our first aim is to understand learn about Hyperbolic pair of pants. For this, we need to recall the concepts of Hyperbolic geometry and we do that in this section.

2 Hyperbolic geometry I

Definition 2.1. *Upper half-plane.* The upper half-plane is $\{z \in \mathbb{C} : \text{Im}(z) > 0\}$. We denote it by $\mathbb{H}$.

We introduce the Riemannian metric

$$\frac{dx^2 + dy^2}{y^2}$$

on $\mathbb{H}$. This metric scales lengths (from the Euclidean one) by a factor of $1/y$ and areas by $1/y^2$. The angles remain same as in the Euclidean case.

Definition 2.2. *Upper half-plane model.* $\mathbb{H}$ equipped with the mentioned Riemannian metric is called the upper half-plane model.

How do we get this metric? One way to see it is (if we already know about Poincare disk model) that we consider the Poincare disk model (D, ρ) where $\rho = \frac{4(dx^2+dy^2)}{(1-x^2-y^2)^2}$ and consider the map $\varphi : D \rightarrow \mathbb{H}$ such that $\varphi(\zeta) = \frac{i(1+\zeta)}{(1-\zeta)}$. Suppose, that we impose the ‘isometry condition’ on φ i.e. φ is a diffeomorphism from D to $\mathbb{H}$ such that for every $P \in D$ and $x, y \in \mathbb{C}$, we get

$$\langle x, y \rangle_P = \langle d\varphi_P(x), d\varphi_P(y) \rangle_{\varphi(P)}^{\sim}$$

where $\langle, \rangle$ is the Poincare disk metric and $\langle, \rangle^{\sim}$ is the induced Riemannian metric on $\mathbb{H}$ that we are trying to find. This metric induced on $\mathbb{H}$ is exactly the Riemannian metric $(dx^2 + dy^2)/y^2$. *And how do we get the Poincare disk metric?* We get it through a similar process involving Hyperboloid model!

We will be extensively using the orientation preserving isometries of $\mathbb{H}$ and so we study it now.

Proposition 2.3. *The orientation preserving isometries, $Is^+(\mathbb{H})$, of $\mathbb{H}$ is exactly the group $PSL(2, \mathbb{R})$.*

Recall: $PSL(2, \mathbb{R}) = \left\{ z \mapsto \frac{az+b}{cz+d} : a, b, c, d \in \mathbb{R} \text{ and } ad - bc = 1 \right\}$.

Proof. We use the fact from group theory that $PSL(2, \mathbb{R})$ is generated by

1. Dilations. $z \mapsto az, a > 0$
2. Translations. $z \mapsto z + a, a \in \mathbb{R}$
3. Single map. $z \mapsto -\frac{1}{z}$

Here, both the domains and the codomains are $\mathbb{H}$. It suffices to show that the metric $|dz|^2/y^2$ is preserved by each of these transformations where $z = x + iy$. The first and the second cases are easy and so we deal with the third case here.

Now, let f be the map $z \mapsto -\frac{1}{z}$. Then, $f'(z) = \frac{1}{z^2}$. So, $dz \mapsto d(-1/z) = \frac{dz}{z^2}$ and hence,

$$|d(-1/z)|^2 = \frac{1}{|z|^4} |dz|^2.$$

Also, $\Im(-1/z) = -\frac{1}{|z|^2}\Im(\bar{z}) = \frac{1}{|z|^2}\Im(z)$.

Combining both the facts, we get that

$$\frac{|d(-1/z)|^2}{\Im(-1/z)^2} = (|dz|^2/|z|^4)/(\Im(z)^2/|z|^4) = \frac{1}{(\Im(z))^2}|dz|^2.$$

So, third transformation preserves the metric. $\square$

Definition 2.4. *Hyperbolic lines. They are either vertical lines or semi-circle with both ends on $\mathbb{R}$. They are also the geodesic lines in $\mathbb{H}$.*

Lemma 2.5. *$PSL(2, \mathbb{R})$ acts transitively on hyperbolic lines in $\mathbb{H}$.*

Proof. Given any two hyperbolic lines l_1 and l_2 , we need to show that $\exists g \in PSL(2, \mathbb{R})$ such that $g(l_1) = l_2$.

Case I: l_1, l_2 are both vertical half-lines.

Proof. Easy. We just use a translation $z \mapsto z + a$.

Case II: At least one of l_1, l_2 is a semi-circle with diameter on $\mathbb{R}$.

Proof. It suffices to take $l_2 = \mathbb{I} = \{z \in \mathbb{H} : \Re(z) = 0\}$ because if g takes l_1 to $\mathbb{I}$, $\tilde{g} \in PSL(2, \mathbb{R})$ takes l_2 to $\mathbb{I}$, then $\tilde{g}^{-1} \circ g \in PSL(2, \mathbb{R})$ does the needed thing.

Now, if l_1 is a half-vertical line, we return to **case I**. If not and if the semi-circle cuts $\mathbb{R}$ at $s < t$ then we consider $z \mapsto \frac{z-t}{z-s}$. Its determinant is $t-s > 0$. So, divide numerator and denominator by $\sqrt{t-s}$ to get the required $PSL(2, \mathbb{R})$ transformation.

What this transformation does is that it maps s to ∞ and t to 0. The hyperbolic line crossing ∞ is $\mathbb{I}$. So, $z \mapsto \frac{z-t}{z-s}$ maps l_1 to $\mathbb{I}$ because elements of $PSL(2, \mathbb{R})$ maps hyperbolic lines to hyperbolic lines. $\square$

We have a stronger result than this. For any $P \in l_1$ not in the endpoints of l_1 , $\exists g \in PSL(2, \mathbb{R})$ such that $g(P) = i$ and g moves l_1 to $\mathbb{I}$. To see this, just note that we can compose the $PSL(2, \mathbb{R})$ element from **case II** with $z \mapsto az$. So,

Lemma 2.6. *$PSL(2, \mathbb{R})$ acts transitively on the pairs of the form (l, P) where l is any hyperbolic line and P is any non-end point on it.*

Proposition 2.7. *Let $\gamma : [0, 1] \rightarrow \mathbb{H}$ be a C^1 -curve such that $\gamma(0) = z_1$ and $\gamma(1) = z_2$. Then, $\text{length}(\gamma) \geq d_{\mathbb{H}}(z_1, z_2)$ with equality iff γ is the monotonic parametrization of $[z_1, z_2] \subseteq l$, where l is the hyperbolic line through z_1 and z_2 .*

Proof. We choose $g \in PSL(2, \mathbb{R})$ such that $g(l) = \mathbb{I}$. So, z_1 maps to some iu and z_2 maps to some iv . Wlog, assume that $u < v \in \mathbb{R}$.

Now,

$$\begin{aligned} \text{length}(\gamma) &= \int_0^1 \frac{1}{y} \sqrt{\dot{x}^2 + \dot{y}^2} dt \\ &\geq \int_0^1 \frac{1}{y} |\dot{y}| dt \\ &\geq \left| \int_0^1 \frac{\dot{y}}{y} dt \right| \\ &= |\log y(1) - \log y(0)| \\ &= \log(v/u). \end{aligned}$$

All the inequalities change into equality when $\gamma = [iu, iv] \subseteq l$ and has monotone parametrisation. This gives the segment of the half line from iu to iv . So, $d_{\mathbb{H}}(z_1, z_2) = \log(v/u)$ and $\text{length}(\gamma) \geq d_{\mathbb{H}}(z_1, z_2)$. $\square$

Theorem 2.8. *If a is a geodesic and p is a point in $\mathbb{H}$, then $\exists$ through p a unique geodesic perpendicular to a .*

Proof. It is easy when a is a half-line. So, we suppose that a is a semi-circle with ends $s < t \in \mathbb{R}$. Now, choose $g \in PSL(2, \mathbb{R})$ such that g translates this semi-circle to a position where $\mathbb{I}$ is orthogonal to it i.e. $\mathbb{I}$ bisects the arc. Let $p' = g(p)$. If p' lies on $\mathbb{I}$, then we take the segment of $\mathbb{I}$ from p' to the point where the semi-circle crosses $\mathbb{I}$.

Otherwise, we draw the geodesic semi-circles from p' to s and t , σ_s and σ_t respectively. Then, they make angles 0 and 2π radians with the semi-circle. If $z \in \mathbb{R}$ is such that $s < z < t$ then let θ_z be the angle made by σ_z with the semi-circle. Noting that $z \mapsto \theta_z$ is continuous where the domain is $[s, t]$ and co-domain is $[0, 2\pi]$, using Intermediate value theorem, $\exists w \in [s, t]$ such that $\theta_w = \pi/2$ and it is the unique one because if there is another, they together would make a triangle with total angle sum $\geq \pi$. We take g^{-1} of this unique geodesic to get the one in the original scenario. $\square$

Theorem 2.9. *If a and b are geodesics with positive distance, then $\exists!$ geodesic perpendicular to a and b .*

Idea of Proof. Take $g \in PSL(2, \mathbb{R})$ such that $g(a) = \mathbb{I}$. $g(b)$ can't be a vertical line because g preserves the distance and we had assumed that a and b have a positive distance. So, $g(b)$ is a semi-circle with endpoints in $\mathbb{R}$: some $s < t$. Let σ_s be the semi-circle centered at O (so that it is orthogonal to $\mathbb{I}$) and passing through s . σ_t be a similar one. Then, again by a continuity argument and Intermediate value theorem, $\exists$ a geodesic perpendicular to a and b . It is the unique one due to a similar reason as earlier (geodesic quadrilateral). $\square$

Where do we use these last two theorems?

We use it to prove Theorem 3.14 and to show the existence of a hyperbolic hexagon with consecutive alternating side lengths specified in Theorem 5.1. This in turn helps us to show in Section 5.2 the existence of the hyperbolic pair of pants with closed boundary geodesic lengths specified.

To talk about the hyperbolic pair of pants, we first need to study how we can form hyperbolic surfaces using the hyperbolic plane. We do it in this section. The following chapter is based on [1] chapter 1.

3 Constructing hyperbolic surfaces

3.1 Hyperbolic atlas

Definition 3.1. *Riemannian surface.* A smooth, orientable 2-dimensional manifold possibly with a piecewise smooth boundary equipped with a Riemannian metric.

We can see a hyperbolic surface in at least two different ways:

- A complete Riemannian surface with constant curvature of -1.
- A connected surface $\mathcal{S}$ equipped with a hyperbolic structure that becomes complete with respect to the induced metric.

We will briefly consider the first approach when we talk about Riemann surfaces in Section 6. For now, we stick with the second approach.

So, what is a hyperbolic structure on $\mathcal{S}$ then?

It is the maximal collection of coordinate charts $\mathcal{A} = \{(U_i, \varphi_i)\}_i$ such that

- $\varphi_i : U_i \rightarrow \varphi_i(U_i) \subseteq \mathbb{H}$ is a homeomorphism.
- the sets U_i cover $\mathcal{S}$.
- if $(\varphi, U), (\varphi', U') \in \mathcal{A}$, then considering each connected component V of $U \cap U'$, $\exists$ an isometry $m \in PSL(2, \mathbb{R})$ such that $\varphi \circ \varphi'^{-1} : \varphi'(V) \rightarrow \varphi(V)$ coincides with m on $\varphi'(V)$.

Hyperbolic atlas for surfaces with boundary.

To define this, we must clarify what the neighborhoods of boundary points look like. They can't be the usual 'open set containing the point inside'. For this, we introduce the concept of circle sector.

Definition 3.2. *Circle sector.* Consider the set of points $V = \{(\rho, \sigma) \in \mathbb{H} : 0 < \rho < \delta, 0 \leq \sigma \leq v\} \cup \{p_0\}$. We call this a circle sector of radius δ and angle v based at p_0 . Here, (ρ, σ) denote polar coordinates in the hyperbolic plane.

What is it geometrically?

Take two geodesic segments of length δ emanating from p_0 and their tangents making an angle of v . The area between the arms (excluding the points at a distance of exactly δ units from p_0) is V .

Definition 3.3. *Vertex.* $p \in \partial\mathcal{S}$ is a vertex if it is the endpoint of 2 sides meeting at p .

Definition 3.4. *Boundary point.* $p \in \partial\mathcal{S}$ is a boundary point if it is not a vertex.

Definition 3.5. *Hyperbolic structure (boundary case).*

It is the maximal collection of coordinate charts $\mathcal{A} = \{(U_i, \varphi_i)\}_i$ such that:

1) For each $p \in \mathcal{S}$, $\exists$ a coordinate chart $(\varphi, U) \in \mathcal{A}$ with $p \in U$ such that $\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{H}$ is a homeomorphism and $\varphi(U)$ is

- an open disc centered at $\varphi(p)$ if p is an interior point.

- a circle sector at $\varphi(p)$ with angle $\leq \pi$ if p is a vertex
- a half-disk centered at $\varphi(p)$ if p is an ordinary boundary point.

2) The sets U_i cover $\mathcal{S}$.

3) Suppose, $(\varphi, U), (\varphi', U') \in \mathcal{A}$, then considering each connected component V of $U \cap U'$, $\exists$ an isometry $m \in PSL(2, \mathbb{R})$ such that $\varphi \circ \varphi'^{-1} : \varphi'(V) \rightarrow \varphi(V)$ coincides with m on $\varphi'(V)$.

Definition 3.6. *Complete hyperbolic structure. If metric induced by the hyperbolic structure on $\mathcal{S}$ is complete then we call it complete hyperbolic structure.*

Definition 3.7. *Hyperbolic surface. A surface $\mathcal{S}$ equipped with a complete hyperbolic structure.*

3.2 Induced metric

What is the metric induced by any hyperbolic structure?

In Section 1, we introduced the metric g , referring to it as the descended metric. Let us now describe precisely what this means in the case of a surface $\mathcal{S}$. The case of the pair of pants requires additional care, which we will discuss later.

Let $c : [0, 1] \rightarrow \mathcal{S}$ be a curve on $\mathcal{S}$. First, we find a sub-division $a = t_0 < t_1 < \dots < t_n = b$ so that $c[t_i, t_{i+1}] \subseteq U_i$ where U_i is the domain of some coordinate chart (U_i, φ_i) . Define:

$$l(c) := \sum_{i=0}^{n-1} l_{\mathbb{H}}(\varphi_i \circ c|_{[t_i, t_{i+1}]}) ,$$

where $l_{\mathbb{H}}(\cdot)$ measures the length in the hyperbolic plane and $l(\cdot)$ measures the length of a curve in $\mathcal{S}$.

Remark 3.8. *One can show that $l(c)$ does not depend on the chosen subdivision of $[0, 1]$. Moreover, if d and d' are two locally isometric metrics on $\mathcal{S}$, both locally isometric to $d_{\mathbb{H}}$, then the length of c is the same with respect to both.*

This definition of length yields the following Riemannian metric on $\mathcal{S}$, called the metric induced by the hyperbolic structure:

$$d(x, y) = \inf\{l(c) | c : [0, 1] \rightarrow \mathcal{S}, c(0) = x, c(1) = y\}$$

We will equip $\mathcal{S}$ with this metric d whenever we refer to the metric induced by the hyperbolic structure.

The case of the pair of pants requires additional care, since its hyperbolic structure is obtained by gluing two hyperbolic surfaces. We now describe the pasting procedure.

3.3 Pasting

Let $\mathcal{S}, \mathcal{S}'$ be two disjoint convex polygons in $\mathbb{H}$. Suppose γ is a side of $\mathcal{S}$, γ' is a side of $\mathcal{S}'$ and they are of the same length and have opposite boundary orientation.

Parametrize $\gamma : [0, 1] \rightarrow \mathcal{S}$ and $\gamma' : [0, 1] \rightarrow \mathcal{S}'$ with the same speed and opposite boundary orientations.

Consider the quotient

$$F = (\mathcal{S} \sqcup \mathcal{S}') / \sim,$$

where $\sim$ is the equivalence relation that identifies $\gamma(t)$ with $\gamma'(t)$ for each $t \in [0, 1]$, and leaves every other point unchanged. Then F is the oriented quotient space obtained by gluing $\mathcal{S}$ and $\mathcal{S}'$ along γ and γ' . More general surfaces can be glued together in the same way.

The hyperbolic pair of pants $P_{(a,b,c)}$ is obtained by gluing two copies of a geodesic hexagon. This is shown in Section 5.2. But are these hexagons hyperbolic surfaces? This is an important question because the condition that guarantees pasting gives a hyperbolic surface Theorem 3.12 assumes that surfaces being glued are hyperbolic surfaces. We answer this now.

3.4 Constructing a hyperbolic structure from a fundamental domain

Suppose we are given a surface with fundamental domain $\mathcal{R}$, and we wish to show that it is a hyperbolic surface. *How can we construct its hyperbolic structure?*

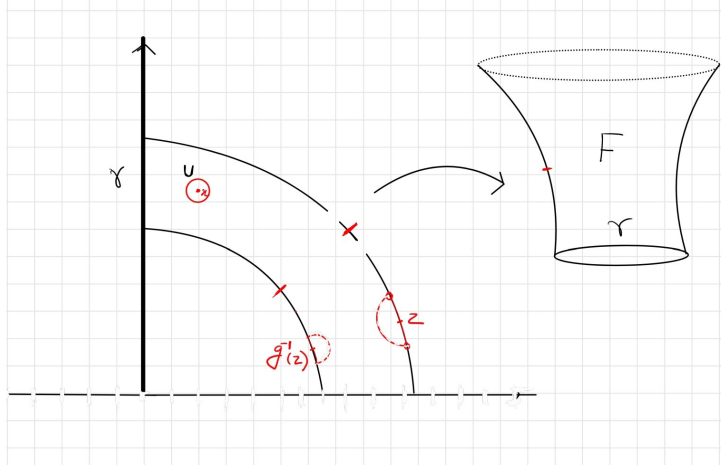


Figure 3: The fundamental domain $\mathcal{R}$ for the funnel F . A and B are the areas inside the two semicircles.

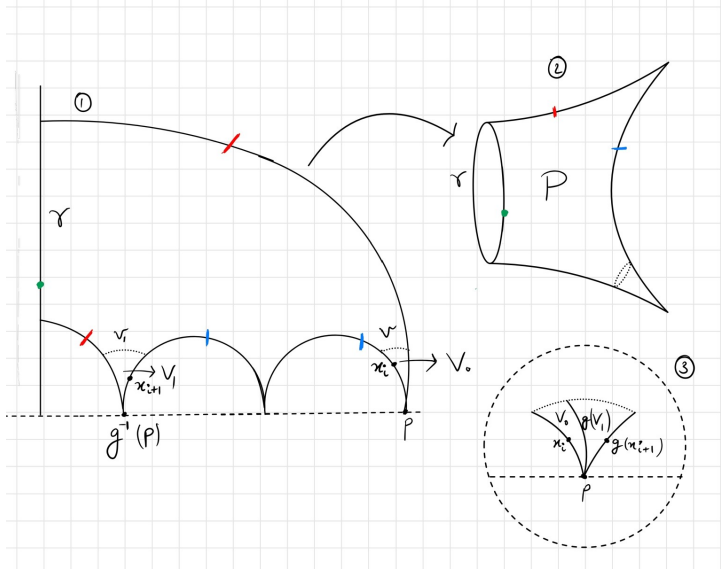


Figure 4: (1) shows the fundamental domain of the surface P . (3) shows that $\varphi(V_1)$ lies just to the right of $V_0 = \varphi(V_0)$, and together they form a circle sector in $\mathbb{H}$.

Construction. Let G be the set of the side-pairing isometries for the fundamental domain $\mathcal{R}$. Define an equivalence relation $\sim$ on $\overline{\mathcal{R}}$ by declaring $x \sim y$ if there exists $g \in G$ such that

$x = g(y)$. The quotient $\mathcal{R}/\sim$ is a surface, though not yet a hyperbolic one. We equip it with the quotient topology induced by $\sim$. Points in $\text{int}(\mathcal{R})$ are one-point equivalence classes. Points in $\text{int}(\partial\mathcal{R})$ are in 2-point equivalence classes. Here, int stands for interior. If x is a vertex, then its equivalence class contains all the vertices of its vertex cycle.

Definition 3.9. *Vertex cycle.* The set of all vertices in the surfaces that together map to a single point in the surface F after pasting form a vertex cycle.

View $\mathcal{R}$ as embedded in $\mathbb{H}$ via the inclusion map $\iota : \mathcal{R} \hookrightarrow \mathbb{H}$.

1. If $x \in \text{int}(\mathcal{R})$, take $\varphi = \text{Id}$ and U an open ball inside $\mathcal{R}$.
2. Suppose $z \in s$, where s is a side of $\mathcal{R}$, and assume z is not a vertex. Choose $r > 0$ such that $B_r(z) \cap \mathcal{R}$ is an open half-disk centered at z and bounded by s . Then $B_r(g^{-1}(z)) \cap \mathcal{R}$ is also an open half-disk centered at $g^{-1}(z)$, where g is the side-pairing element associated with s . The construction works because $\mathcal{R}$ is convex and z is not a vertex.

Set $A = (B_r(z) \cap \mathcal{R}) \cup (B_r(z) \cap s)$. This is the open half disk plus the interior of the diameter.

Similarly, let $B = (B_r(g^{-1}(z)) \cap \mathcal{R}) \cup (B_r(g^{-1}(z)) \cap g^{-1}(s))$. For an example, see Figure 3.

In the quotient $\mathcal{R}/\sim$, a point $x \in A \cap s$ is identified with $y \in B \cap g^{-1}(s)$ whenever $x = g(y)$. Thus A and B are glued along $s \cap A$ and $g^{-1}(s) \cap B$. Let U be the projection of $A \cup B$ to $\mathcal{R}/\sim$. Then U is an open ball in $\mathcal{R}/\sim$, forming the coordinate chart (U, φ) . Define φ by

$$\varphi(x) = \begin{cases} x, & x \in A, \\ g(x), & x \in B. \end{cases}$$

Then φ is a homeomorphism from $A \cup B$ onto U . Please refer Figure 4 for the action of φ .

3. Suppose $p \in \mathcal{R}$ is a vertex. Choose $r > 0$ small enough such that $B_r(p) \cap \mathcal{R}$ is a circle sector with center p and angle v . Let the vertex cycle containing p be $\{p_1, \dots, p_k, p\}$, with associated isometries $g_1, \dots, g_k$. Then the vertex cycle consists of the points $g_1^{-1}(p), \dots, g_k^{-1}(p), p$.

Let the associated circle sectors be $V_1, \dots, V_k, V_0$ with angles $v_1, \dots, v_k, v$. We suppose that the cycle is numbered in such a way that the left hand side of V_i is identified with the right-hand side of V_{i+1} . The circle sectors may lie on different copies of $\mathcal{R} : \mathcal{R}_1, \dots, \mathcal{R}_k$.

Take $V := V_1 + \dots + V_0 \text{ mod } (*)$ (the pasting condition on circle sectors induced by $\sim$) and φ on V as follows:

- if $x \in V_i$ with $i \neq 0$, then $\varphi(x) = g_i(x)$
- if $x \in V_0$, then $\varphi(x) = x$

Then, through elements of the form (V, φ) , we get a hyperbolic structure $\mathcal{A}$ for $\mathcal{R} \cup \mathcal{R}_1 \cup \dots \cup \mathcal{R}_k / \sim$.

First, we need to check that φ is well-defined. Take x_i on the left-side of V_i and x_{i+1} on the right-side of V_{i+1} such that $\text{dist}(p_{i+1}, x_{i+1}) = \text{dist}(p_i, x_i) = r < \epsilon$.

Now, φ is well-defined only if $\varphi(x_i) = \varphi(x_{i+1})$. Now, $\varphi(x_i)$ lies on the left side of the $g_i(V_i)$ at a distance of r units from the vertex of the circle sector of $g_i(V_i)$. Call it y_i . Similarly, $\varphi_{i+1}(x_{i+1})$ lies on the right side of $g_{i+1}(V_{i+1})$ at a distance of r units from the vertex of the circle sector $g_{i+1}(V_{i+1})$. Call it y_{i+1} .

Since x_i and x_{i+1} are identified under the equivalence relation, we have $y_i = y_{i+1}$. Thus the map is well-defined. $\mathcal{A}$ is then a hyperbolic structure if third condition of Definition 3.5 is satisfied. We skip it and check that in the proof of Claim 1 in Theorem 3.11 (similar last part). $\square$

Remark 3.10. We see in Figure 3 that A and B appear to be of different sizes. It appears like this because we are trying to visualize areas in hyperbolic plane projected on the Euclidean plane. Also, not every element of $PSL(2, \mathbb{R})$ would translate the inner arc to the outer arc. These elements are called hyperbolic. Similarly, in Figure 4, the translation of the first semi-circle to the second one (both marked in blue) is done by elements called parabolic. We will study more about these elements in the next chapter.

Theorem 3.11. Take $\mathcal{S}$ to be a hyperbolic surface, $\Gamma < Is^+(\mathcal{S})$ such that it acts freely and properly discontinuously on $\mathcal{S}$. Then, $\mathcal{S}/\Gamma$ carries a uniquely determined complete hyperbolic structure such that $\pi : \mathcal{S} \rightarrow \mathcal{S}/\Gamma$ is a local isometry.

Proof. Let $\mathcal{A}$ be a hyperbolic atlas of $\mathcal{S}$. So, we need to construct a hyperbolic atlas $\mathcal{A}'$ for $\mathcal{S}/\Gamma$ and show that it is a complete structure.

Take $q \in \mathcal{S}/\Gamma$. We need to find a coordinate pair (V, φ') around each such q satisfying all 3 conditions in Section 3.1.

Now, let $\tilde{q} \in \pi^{-1}(q)$. If q is an ordinary boundary point, then $\tilde{q}$ is either an ordinary boundary point or a vertex point. Let $V \subseteq \mathcal{S}/\Gamma$ be an elementary neighborhood around q . This is possible because π is a covering map by Lemma A.20. So, $\pi^{-1}(V)$ is a disjoint union of open sets $\cup_{i \in I} V_i$ s.t. $\pi|_{V_i} : V_i \rightarrow V$ is a homeomorphism. Moreover, for any $i, j \in I$, there is a deck transformation with $\gamma \in \Gamma$ s.t. $\gamma(V_i) = V_j$.

Let i be the index s.t. $\tilde{q} \in V_i$. Let $(U_i, \varphi_i) \in \mathcal{A}$ with radius $\delta > 0$ small enough so that $U_i \subseteq V_i$ and such that $\varphi_i(U_i)$ is a open disk/half-disk/circle sector centered at $\varphi_i(\tilde{q})$ depending on whether $\tilde{q}$ is an interior point/ordinary boundary point/vertex point.

Why is this possible? This is possible because $\mathcal{S}$ is equipped with the maximal hyperbolic atlas.

Now, consider the chart $(\pi(U_i), \varphi_i \circ \pi|_{U_i}^{-1})$ and form the maximal atlas $\mathcal{A}'$ with such elements.

Had q been a vertex point, we would have restricted $\tilde{q}$ to the vertex point case. If q had been an interior point, then $\tilde{q}$ could have been interior point, boundary point or vertex point. We deal with the case for the surface $\mathcal{S}$ without boundary i.e. all points are interior points, for simplicity.

Claim 1: Condition 3 in Definition 3.5 is satisfied.

Proof of claim 1. Consider $(\pi(U), \varphi \circ (\pi|_U)^{-1})$ and $(\pi(U'), \varphi' \circ (\pi|_{U'})^{-1}) \in \mathcal{A}'$. Then, we need to show that for each connected component W of $\pi(U) \cap \pi(U')$, $\exists$ an isometry $m \in PSL(2, \mathbb{R})$ such that

$$\varphi' \circ (\pi|_{U'})^{-1} \circ (\pi|_U) \circ \varphi^{-1}$$

coincides with m on $\varphi \circ (\pi|_U)^{-1}(W)$.

We want to show that the map is an isometry from $\varphi(B)$ to $\varphi'(B')$ where $B' = (\pi|_{U'})^{-1}(W)$ and $B = (\pi|_U)^{-1}(W)$.

Now, $\exists$ a deck transformation $\gamma \in \Gamma$ s.t. $\gamma(B) = B'$ because they are the lifts of the same image downstairs which is a subset of some elementary neighborhood. Restrict φ' to the domain B' and φ to the domain B . Then, $\varphi' \circ \gamma^{-1} \circ \varphi^{-1}$ coincides with some $m \in PSL(2, \mathbb{R})$ restricted to $\varphi(B)$ as it is a composition of isometries. So, $\varphi' = m \circ \varphi \circ \gamma$ on B' .

So, $\varphi' \circ (\pi|_{U'})^{-1} \circ (\pi|_U) \circ \varphi^{-1} = m \circ \varphi \circ \gamma \circ (\pi|_{U'})^{-1} \circ (\pi|_U) \circ \varphi^{-1} = m$ from $\varphi(B)$ to $\varphi'(B')$.

Claim 2: $\pi : \mathcal{S} \rightarrow \mathcal{S}/\Gamma$ is a local isometry.

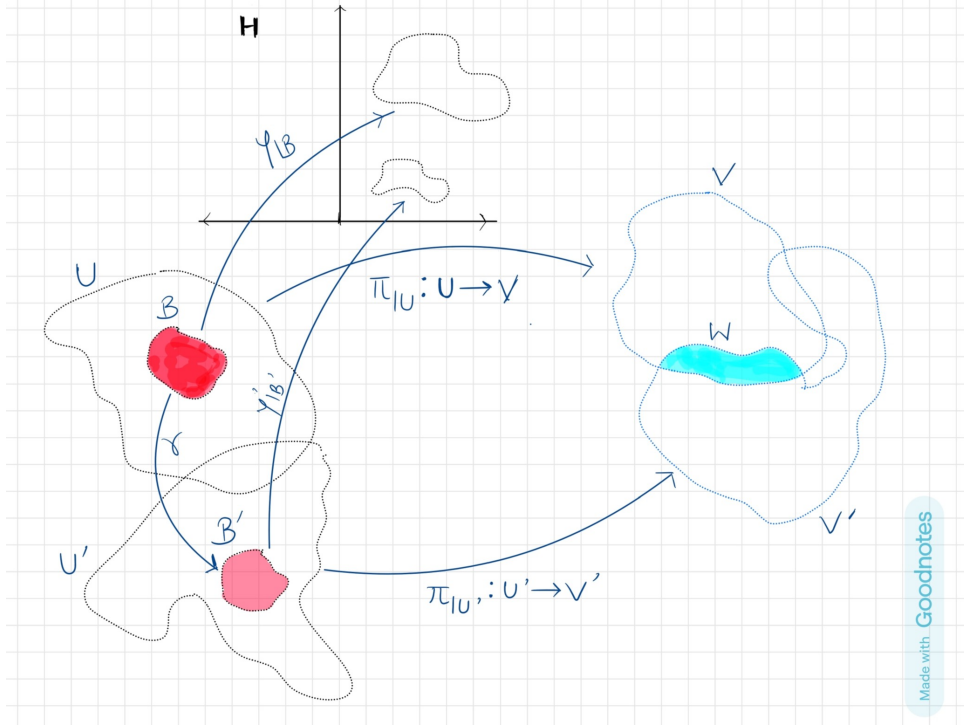


Figure 5: $B = \pi_U^{-1}(W)$ and $B' = \pi_{U'}^{-1}(W)$.

Consider the metric induced on $\mathcal{S}/\Gamma$ according to 3.2. Now, we know that V is essentially the same as V_i and so the distances between two points in the open sets are same. For every $p \in \mathcal{S}/\Gamma$, we can do this and so it is a local isometry.

Claim 3: The constructed hyperbolic structure is complete.

Now, π is a covering map and so completeness follows by property of Riemannian covering map. $\square$

3.5 A counter-example

Not every pasting of hyperbolic surfaces gives us a hyperbolic surface. We had mentioned that the pair of pants is obtained through pasting and we need to know if it is a hyperbolic surface or not. However, before that, we show that gluing need not yield a hyperbolic surface:

Consider $\mathbb{H}$ and the geodesics $\alpha(t) = 1 + ie^t$ and $\beta(t) = 2 + ie^t$ with $t \in \mathbb{R}$. The closed strip S between α and β is a surface with boundary, endowed with the metric induced from the hyperbolic plane.

Now, our goal is to show that if we paste α and β in a certain way, the resultant surface is not complete and hence not hyperbolic.

For this, consider the isometry $m : z \mapsto 2z$. We paste the side α with β using m i.e. the pasting condition is

$$\alpha(t) = 1 + ie^t \sim 2 + i2e^t = \beta(t + \log 2), \quad t \in \mathbb{R} \quad (*)$$

Now, let $F = S \bmod (*)$.

Claim: $\exists$ a Cauchy sequence (p_n) in F such that it doesn't converge.

Proof. Take $a_n = 1 + i2^n$ and $b_n = 2 + i2^{n+1}$. Then, (a_n) and (b_n) project to the same points in F by $(*)$. Call it (p_n) .

Since the projection $S \rightarrow F$ is distance non-increasing and identifies these pairs,

$$d_F(p_{n-1}, p_n) \leq \min\{d_{\mathbb{H}}(a_n, a_{n-1}), d_{\mathbb{H}}(a_n, b_{n-1}), d_{\mathbb{H}}(b_n, b_{n-1}), d_{\mathbb{H}}(a_{n-1}, b_n)\}.$$

Also, we have

$$d_{\mathbb{H}}(a_n, b_{n-1}) \leq \frac{1}{2^n},$$

so (p_n) is Cauchy in F . Hence F is not complete, so it cannot carry a complete hyperbolic structure. $\square$

3.6 A sufficient condition

Consider the pairwise disjoint hyperbolic surfaces $\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_m$ and let $\gamma_1, \gamma'_1, \gamma_2, \gamma'_2, \dots, \gamma_n, \gamma'_n$ (the set of sides taking part in the pasting process) be the pairwise distinct sides of $\mathcal{S} := \mathcal{S}_1 \cup \dots \cup \mathcal{S}_m$ such that for each i , the parametrizations are $\gamma_i : I = [0, 1] \rightarrow \mathcal{S}$ and $\gamma'_i : I \rightarrow \mathcal{S}$ with the same constant speed. When the sides are smooth closed geodesics, we take $\mathbb{S}^1$ instead of I . Pasting is done as defined in Section 3.3.

So, what guarantees that pasting pairwise disjoint hyperbolic surfaces $\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_m$ results in a hyperbolic surface?

The following theorem ensures this.

Theorem 3.12. *1. For any interior point in the surface $F = \mathcal{S}_1 + \mathcal{S}_2 + \dots + \mathcal{S}_m \bmod (*)$, the sum of the interior angles at the vertices in the vertex cycle of this point is 2π .*

2. For any boundary point in the surface F , the sum of the interior angles at the vertices in the vertex cycle of this point is $\leq \pi$.

Then, $\exists!$ hyperbolic structure such that when F equipped with this, the projection map $\sigma : \mathcal{S}_1 \cup \dots \cup \mathcal{S}_m \rightarrow F$ is a local isometry.

Suppose, the following condition is also satisfied:

3. F is connected and any pair of non-adjacent sides in the list $\gamma_1, \gamma'_1, \gamma_2, \gamma'_2, \dots, \gamma_n, \gamma'_n$ lying on the same surface has positive distance.

Then, the above mentioned hyperbolic structure of F is complete as well.

We now prove only the first part here.

Proof. Let $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_m$ be the hyperbolic atlases of the surfaces $\mathcal{S}_1, \dots, \mathcal{S}_m$.

Case 1: Let $p \in F$ be a vertex point and $p_1, \dots, p_r$ be the vertices in the vertex cycle that map to p under σ .

Let the corresponding interior angles made by them be $v_1, \dots, v_r$ whose sum is $v \leq \pi$.

Since the surfaces are pairwise disjoint, taking $\epsilon > 0$ small enough, we can ensure that the neighborhoods

$$U_i = \{x \in \mathcal{S}_1 \cup \dots \cup \mathcal{S}_m \mid \text{dist}(x, p_i) < \epsilon\}$$

are pairwise disjoint for $i \in \{1, 2, \dots, r\}$. We arrange the U_i 's in a right to left fashion i.e. while pasting left side of U_i will be identified with right side of U_{i+1} for $i \in \{1, 2, \dots, r-1\}$.

Choosing ϵ small enough, we can ensure that $(U_i, \varphi_i) \in \mathcal{A}_1 \cup \dots \cup \mathcal{A}_m$ such that $\varphi_i(U_i)$ is a circle sector V_i of radius ϵ and angle v_i .

We observe that if we find isometries in $PSL(2, \mathbb{R}) : m_1, m_2, \dots, m_r$ of $\mathbb{H}$ such that all the circle sectors V_i are brought together in the sense of right to left fashion, then we get that

$V := m_1(V_1) \cup \dots \cup m_r(V_r)$ is a circle sector of angle v . The case where p is an ordinary boundary point or an interior point are similar and easier. We will be working for **Case 1**.

Claim 2: Take $U := U_1 + \dots + U_r \bmod (*)$ and $\varphi := m_1 \circ \varphi_1 + \dots + m_r \circ \varphi_r \bmod (*)$ where $(*)$ is the equivalence relation that identifies sides in the right to left fashion. Then, through elements of the form (U, φ) , we get a hyperbolic structure $\mathcal{A}$ for F .

Firstly, we need to check that the φ is well-defined. Take x_i on the left-side of U_i and x_{i+1} on the right-side of U_{i+1} such that $\text{dist}(p_{i+1}, x_{i+1}) = \text{dist}(p_i, x_i) = r < \epsilon$.

Now, φ is well-defined only if $\varphi(x_i) = \varphi(x_{i+1})$. As homeomorphism carries boundary to boundary and φ_i is also an orientation preserving isometry locally, so $\varphi_i(x_i)$ lies on the left side of the V_i at a distance of r units from the vertex of the circle sector of V_i . Call it y_i . Similarly, $\varphi_{i+1}(x_{i+1})$ lies on the right side of V_{i+1} at a distance of r units from the vertex of the circle sector of V_{i+1} . Call it y_{i+1} .

Now, $m_i(y_i) = m_{i+1}(y_{i+1})$ as y_i, y_{i+1} go to the same point under $(*)$. So, the map is well-defined.

We now check that we indeed get a hyperbolic atlas.

Let $(U, \varphi), (\tilde{U}, \tilde{\varphi}) \in \mathcal{A}$.

Claim 3: For each connected component W of $U \cap \tilde{U}$, $\exists$ an isometry $m \in PSL(2, \mathbb{R})$ s.t. $\varphi \circ \tilde{\varphi}^{-1}$ coincides with m on $\tilde{\varphi}(W)$.

Proof. Similar to Claim 1 in Theorem 3.11.

3.7 Perpendiculars.

Definition 3.13. *Gliding homotopy.* Let A, B be closed connected subsets and $c : [0, 1] \rightarrow \mathcal{S}$, $\gamma : [0, 1] \rightarrow \mathcal{S}$ be paths such that $c(0), \gamma(0) \in A$ and $c(1), \gamma(1) \in B$. c is homotopic to γ with endpoints gliding on A and B if $\exists$ continuous $H : [0, 1] \times [0, 1] \rightarrow \mathcal{S}$ with $H(0, t) = c(t)$, $H(1, t) = \gamma(t)$, $H(s, 0) \in A$, $H(s, 1) \in B \forall 0 \leq s \leq 1$.

Theorem 3.14. Let $\mathcal{S}$ be a hyperbolic surface and let $c : [0, 1] \rightarrow \mathcal{S}$ be a path with endpoints on disjoint closed boundary geodesics A and B . In the homotopy class of c with endpoints gliding on A and B , $\exists$ a unique geodesic γ . At its endpoints, γ meets A and B perpendicularly. All other points of γ lie in the interior of $\mathcal{S}$.

There are two methods to prove this: using Arzela-ascoli theorem or using Semmler's algorithm and Cheeger-Ebin theorem.

Theorem 3.15. *Arzela-Ascoli.* Let X be a separable metric space and Z be a compact metric space. Suppose, (γ_n) is a sequence of L -Lipschitzian mappings for some $L > 0$. Then, $\exists$ a subsequence (γ_{n_k}) such that it converges uniformly on compact sets of X to an L -lipschitzian mapping $\gamma : X \rightarrow Z$.

In this theorem, we can use a compact Riemannian manifold $\mathcal{M}$ as well in place of Z . We use any compact Hyperbolic surface $\mathcal{S}$.

Instead of giving a proof of Theorem 3.14, we prove a similar theorem (easier to prove) which can be adapted for Theorem 3.14.

Theorem 3.16. *Non-trivial free homotopy classes of paths on any closed compact hyperbolic-surface $\mathcal{S}$ contain unique geodesics.*

Claim 1: Every non-trivial free homotopy class of curves on $\mathcal{S}$ contains at least one closed geodesic.

Proof of claim 1. Take some curve $\alpha : \mathbb{S}^1 \rightarrow \mathcal{S}$ and suppose

$$C := \{\eta : \mathbb{S}^1 \rightarrow \mathcal{S} \mid \eta \text{ is freely homotopic to } \alpha\}$$

is non-trivial. Let $L := \inf_{\gamma' \in C} l(\gamma')$. Consider a sequence (γ_n) so that $l(\gamma_n) \rightarrow L$. It follows from the Arzela-Ascoli theorem that $\exists$ a subsequence (γ_{n_k}) such that it converges uniformly on $\mathbb{S}^1$ to some $\gamma : \mathbb{S}^1 \rightarrow \mathcal{S}$ with $l(\gamma) = L$. Since, this is the length minimizing curve in C and $L \neq 0$ so, γ must be a geodesic. $\square$

Claim 2: $\gamma \in C$.

Proof idea of claim 2. We lift α by the covering map $\pi : \mathbb{H} \rightarrow \mathcal{S}$ to some $\tilde{\alpha}$.

Now, $\mathcal{S} = \mathbb{H}/\Gamma$ by corollary 6.6 for Γ acting properly discontinuously and freely. Γ is also its fundamental group. Let $g \in \Gamma$ be the based homotopy class of α . g will be a hyperbolic element. Denote by A_g the axis (defined in Section 4.4.1) of g in $\mathbb{H}$. $\langle g \rangle$ stabilizes $\tilde{\alpha}$. Moreover, the lift of γ (in Claim 1) is A_g and $\tilde{\alpha}$ lies in the neighborhood of A_g with same endpoints. We create a homotopy from $\tilde{\alpha}$ to A_g as follows:

From each point of $\tilde{\alpha}$, we draw the unique geodesic segment cutting A_g orthogonally. The homotopy $\tilde{H}$ moves the points of $\tilde{\alpha}$ continuously along these segments until we get A_g . We project this entire free homotopy downstairs to get a free homotopy H from α to γ . $\square$

Claim 3: This geodesic is unique.

Proof of Claim 3. Consider such a free homotopy $H : [0, 1] \times \mathbb{S}^1 \rightarrow \mathcal{S}$ from α to γ' (another geodesic in C).

Due to the compactness of $[0, 1] \times \mathbb{S}^1$, $\exists$ a constant $K \geq 0$ so that no point of α is moved a distance greater than K by H . By lifting the entire free homotopy, we get a free homotopy between the lifts $\tilde{\alpha}$ and $\tilde{\gamma}'$ and the points of $\tilde{\alpha}$ are moved by at most K units. If $\tilde{\gamma}', \tilde{\alpha}$ have at least one pair of distinct endpoints, then this is not possible. Hence, both the endpoints are same and so γ, γ' coincide. $\square$

Remark 3.17. *This result can be extended to non-compact hyperbolic surface with boundary. To deal with non-compactness, we restrict C to contain only essential curves.*

What if we want to generalise A and B in Theorem 3.14? We can relax it easily to the case where A, B are admissible subsets i.e. compact connected subsets of $\partial\mathcal{S}$:

Theorem 3.18. *Let $\mathcal{S}$ be a hyperbolic surface, A, B admissible subsets and $c : [0, 1] \rightarrow \mathcal{S}$ be a path with $c(0) \in A$ and $c(1) \in B$. Then,*

- *In the gliding homotopy class of c , $\exists$ a length minimizing geodesic γ which is either contained in $\partial\mathcal{S}$ or meets $\partial\mathcal{S}$ at most at its endpoints.*
- *Further, if A, B are from the two options: smooth closed boundary geodesic or a side of $\mathcal{S}$ meeting adjacent sides with angle $\leq \pi/2$, then γ is either a point or the unique perpendicular geodesic from A to B . If c is simple then γ is also simple.*

The proofs of all the claims are similar except for the last fact that γ is also simple. Please refer to [1] Ch 1 for a proof of this. We are going to use Theorem 3.18 when showing that for every pair of closed boundary geodesics of pair of pants, we have a unique simple perpendicular geodesic and all the three perpendiculars cut the pair of pants into two equal parts!

Before finally constructing the pair of pants, we discuss about the elements of $PSL(2, \mathbb{R})$ and their action.

4 An Interlude to Isometries of $\mathbb{H}$

Understanding $\text{Aut}(\hat{\mathbb{C}}) = PSL(2, \mathbb{C})$ gives a lot of information about $PSL(2, \mathbb{R})$ because $PSL(2, \mathbb{R}) < PSL(2, \mathbb{C})$. Now, consider $\text{Aut}(\hat{\mathbb{C}})$ and let $T \in \text{Aut}(\hat{\mathbb{C}})$.

Definition 4.1. $T, \hat{T}$ are conjugates in $\text{Aut}(\hat{\mathbb{C}})$ if $\exists S \in \text{Aut}(\hat{\mathbb{C}})$ such that $\hat{T} = STS^{-1}$.

Remark 4.2. Why do we need conjugacy classes?

We talk about conjugacy classes of elements because inside each conjugacy class, the elements behave in ‘almost the same’ way. If one element rotates the $\hat{\mathbb{C}}$ by θ then so will do the other elements in the conjugacy class. If one expands $\hat{\mathbb{C}}$ by a positive factor not equal to 1, then so will do the other elements in the conjugacy class.

Lemma 4.3. Consider $\text{Aut}(\hat{\mathbb{C}})$. Suppose that z_0 is a fixed point of T .

1. Then, $S(z_0)$ is a fixed point of $\hat{T}$ as $\hat{T}(S(z_0)) = STS^{-1}(S(z_0)) = S(z_0)$.
2. Conjugacy preserves the trace i.e. $(\text{Tr } T)^2 = (\text{Tr } (STS^{-1}))^2$ using the fact that $\text{Tr}(AB) = \text{Tr}(BA)$ when $A, B \in GL(2, \mathbb{C})$.

Let us explore the $\text{Aut}(\hat{\mathbb{C}})$ in details, study the geometric figures traced out by the orbit paths of the elements and figure out if there is any helpful way to further subgroup them depending on their properties. Initially, we try to subgroup them according to the number of fixed points. For this, we introduce the following lemma:

Lemma 4.4. Let $T \in PSL(2, \mathbb{C})$, $T \neq I$. Then, T has either 1 or 2 fixed points. Also, T has exactly one fixed point iff $(\text{Tr } T)^2 = 4$.

Proof. Take $T : z \rightarrow \frac{az+b}{cz+d}, T \in PSL(2, \mathbb{C})$. Now, $T(z) = z \iff cz^2 + z(d-a) - b = 0 \iff z = \frac{(a-d) \pm \sqrt{(d-a)^2 + 4bc}}{2c}$ if $c \neq 0$. So,

$$z = \frac{(a-d) \pm \sqrt{(a+d)^2 - 4}}{2c}$$

using $ad - bc = 1$. So, there is exactly one fixed point iff $(\text{Tr } T)^2 = (a+d)^2 = 4$ and it has 2 fixed points otherwise (When $c = 0$, there are two fixed points). $\square$

Here, I is the identity element of $PSL(2, \mathbb{C})$ i.e $I = \pm \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

4.1 Classification by number of fixed points

In all cases, we are initially going to study the elements whose fixed points are ∞ or 0 or both. Then, we are going to understand the orbits of general elements using *Coaxial circles*.

1) Parabolic elements

Consider $T \in PSL(2, \mathbb{C})$ with exactly one fixed point z_0 . By Lemma 4.4, $\text{Tr } T = \pm 2, T \neq I$. Using Lemma 4.3, we get that if $\hat{T} = STS^{-1}, S \in PSL(2, \mathbb{C})$ then $T, \hat{T}$ have the same

squared trace, $\hat{T} \neq I$ and so $\hat{T}$ still has exactly one fixed point by Lemma 4.4. So, we choose $S \in PSL(2, \mathbb{C})$ such that $S(z_0) = \infty$. Then, the fixed point of $\hat{T}$ is exactly ∞ by Lemma 4.3. Suppose, we are in this case.

Then, $T(z) = \frac{az+b}{cz+d}$, where $ad - bc = 1$ and such that $T(\infty) = \infty$. Now, $T(\infty) = \infty \implies c = 0$ and $ad = 1 \implies d = 1/a$. Applying $Tr T = \pm 2$, we get that $a = \pm 1$. So, $T(z) = z + b$, $b \in \mathbb{C} \setminus \{0\}$. Hence, T is a simple Euclidean translation in this case.

2) Loxodromic elements

Suppose, T has 2 fixed points z_0 and z_1 . Again, we conjugate by some element $S \in PSL(2, \mathbb{C})$ such that $S(z_0) = 0$ and $S(z_1) = \infty$. To be precise, we choose $S : z \mapsto \frac{z-z_0}{z-z_1}$. Then, $\hat{T}$ has fixed points 0 and ∞ . Suppose, we are in this case.

Then, by using the constraints, we get that $T = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}$ so that $T(z) = a^2 z$. Now, $T \neq I$ by Lemma 4.4. So, $a \neq \pm 1$. Then, $T(z) = \lambda z$ where $\lambda \in \mathbb{C} - \{\pm 1\}$. We call λ the multiplier of T . However, the set of Loxodromic elements is too big and we can further subgroup the elements based on how their orbits look which depends on the value of the multiplier.

4.2 Further classification by the value of the multiplier

We do the following classification of the loxodromic elements:

- *Hyperbolic* if $\lambda > 0$ and $\lambda \neq 1$
- *Elliptic* if $|\lambda| = 1$
- *Loxodromic* otherwise.

Consider the case where the fixed points are simply 0 and ∞ . *Why do we classify like that?*

Case 1: It is because when we have *hyperbolic elements*, the orbit paths consist of radial lines coming from 0. When $\lambda > 1$, the hyperbolic element expands the space along the radial lines as in Figure 6. It contracts the space when $\lambda < 1$.

Case 2: When we have an *elliptic element*, the orbit paths are concentric circles around 0 as in Figure 6. So, elliptic elements rotate the plane.

Case 3: When we have any other Loxodromic element, the orbit paths are spirals.

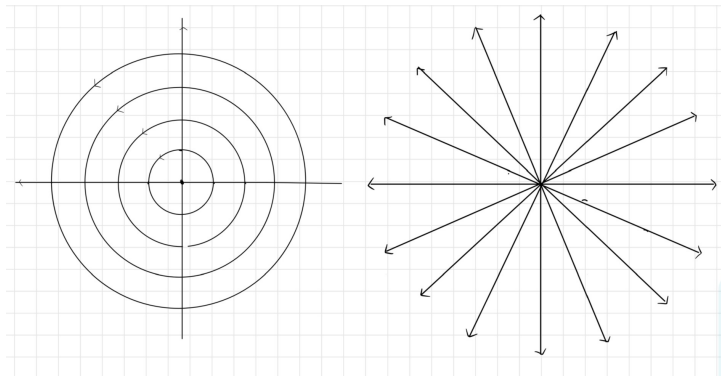


Figure 6: Left: (1) Elliptic. $z \mapsto ze^{i\theta}$. Right: (2) Hyperbolic ($\lambda > 1$). $z \mapsto \lambda z$ expanding $\hat{\mathbb{C}}$ along the radial lines. Fixed points are 0 and ∞ in both.

What about the orbit paths in the more general cases?

Notice that in all the 3 cases (parabolic, elliptic and hyperbolic), the orbit paths are either parallel lines, radial lines or concentric circles. Now, elements of $\text{Aut}(\hat{\mathbb{C}})$ carry circlines to circlines and preserve angles. So, when we conjugate the elements to get more generalised fixed points, the inherent structure formed by them i.e. coaxial circles, is preserved.

4.2.1 Coaxial circles

Coaxial circles are complementary families of circles such that one family of circles is orthogonal to another. This can happen in two ways: **degenerate** and **generic**. In degenerate families of circles, all the circles in the family are tangent to each other at a common point z_0 and the two tangent lines of the two families are orthogonal. In generic case, one family passes through two fixed points z_0^+ and z_0^- while the circles in the complementary family are pairwise disjoint. See Figure 7 for this.

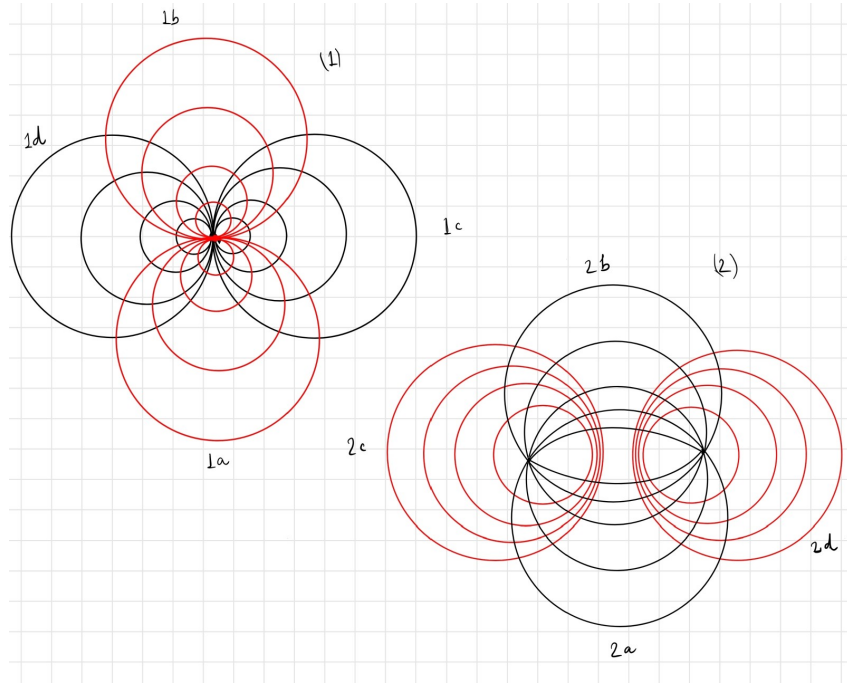


Figure 7: (1): Degenerate families. (1b), (1d) are tangent at a common point, they cut each other orthogonally and tangent lines are at 90° . (2): Generic families. Two families (2b) and (2c) ((1b) and (1d)) form one generic (degenerate) family.

So we look for the general traits of the elements as discussed in **Case 1, 2** and **3** (in section 4.2) in the families of the coaxial circles and choose the families accordingly. *Why?* It is because in Fig 7 and 8, we can see the orbit paths of parabolic, elliptic and hyperbolic elements when the fixed points are 0 and ∞ . Applying $z \mapsto \frac{1}{z-z_0}$ gives (1) in Figure 6 and $z \mapsto \frac{z-z_0^+}{z-z_0^-}$ gives (2) in Figure 6. So,

Parabolic elements: Orbit paths should be parallel circles, passing through a single fixed point z_0 i.e. 1a and 1b in Figure 7 – (1).

Hyperbolic elements: Orbit paths should pass through two fixed points z_0 and z_1 i.e. 2a and 2b in Figure 7 – (2).

Elliptic elements: Orbit paths of elliptic elements should be complementary families to the orbit paths of the hyperbolic elements i.e. 2c and 2d in Figure 7 – (2).

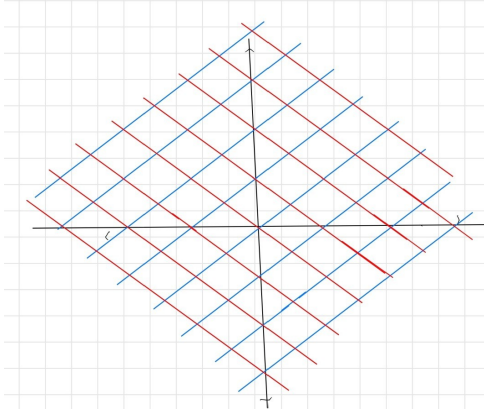


Figure 8: Degenerate coaxial circles family with circles tangent at ∞ .

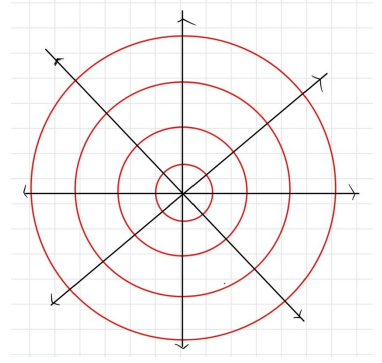


Figure 9: Generic family with parallel circles being cut by circles passing through two fixed points: 0 and ∞ .

4.3 Multiplier and Trace

Consider $T : z \mapsto \lambda z$, $\lambda \in \mathbb{C}$ i.e. a general loxodromic element with two fixed points 0 and ∞ . Now,

$$T = \pm \begin{pmatrix} \sqrt{\lambda} & 0 \\ 0 & \frac{1}{\sqrt{\lambda}} \end{pmatrix}.$$

This form is easy to use but when we have loxodromic elements with non-trivial fixed points, do we need to diagonalise always and find the multiplier to characterise it?

No, we can use just the trace to characterise it.

Before that, there is an abuse of notation that we are making. $\text{Tr } T$ is not well-defined when $T \in \text{PSL}(2, \mathbb{C})$. So, whenever we say that we mean the trace of the pre-images of T in $SL(2, \mathbb{C})$.

Set $l = \log \lambda$. If $\lambda = re^{i\theta}$, then $l = \log r + i(\theta + 2k\pi)$, $k \in \mathbb{Z}$. Then, $e^l = \lambda$. So, $\text{Tr } T = \pm(e^{l/2} + e^{-l/2}) = \pm 2 \cosh(l/2)$. We take $l/2 = \frac{1}{2} \log r + i(\theta/2 + k\pi)$, $k \in \mathbb{Z}$. We will use this to go from λ to l and then to $\text{Tr } T$ giving us iff conditions between trace and multiplier:

Proposition 4.5. *Let $T \in SL(2, \mathbb{C})$. Then,*

1. T is parabolic or $I \iff \text{Tr } T = \pm 2$.
2. T is hyperbolic $\iff \text{Tr } T \in (-\infty, -2) \cup (2, \infty)$.
3. T is elliptic $\iff \text{Tr } T \in (-2, 2)$
4. T is loxodromic $\iff \text{Tr } T$ is not in $\mathbb{R}$

Proof. 1) Using the relation between $\text{Tr } T$ and λ , it is fairly easy to conclude that $\text{Tr } T = \pm 2 \iff T = I$ if $T : z \mapsto \lambda z$. For parabolic, we know the result from section 2.1. For the orbits paths of a general parabolic see Figure 10 - (2).

2) For general hyperbolic elements after conjugation, we have $\lambda > 0, \lambda \neq 1$. This happens by the above calculation iff $r \neq 0, 1$ and $\theta = 2\pi ni$ ($n \in \mathbb{Z}$) $\iff l \in \mathbb{R}_* + 2\pi ki$ ($k \in \mathbb{Z}$) $\iff \text{Tr } T = \pm 2 \cosh(\frac{\mathbb{R}_* + 2\pi ki}{2}) = \pm 2 \cosh(\mathbb{R}_* + \pi ki) > 2$ or < -2 . For the orbits paths of a general hyperbolic see Figure 10 - (3).

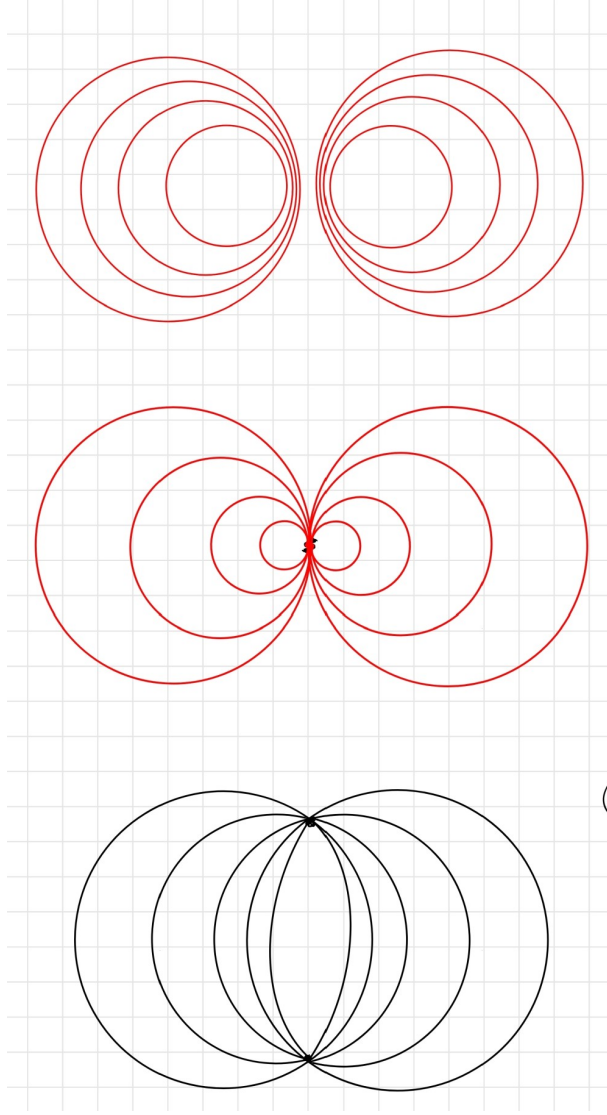


Figure 10: 1) Orbit paths of an elliptic. The fixed points lie inside the two small circles. 2) Orbit paths of a parabolic. Tangent at the fixed point z_0 . 3) Orbit paths of a hyperbolic. Intersect at the fixed points z_0^+, z_0^- .

3) For general elliptic elements, we have $|\lambda| = 1, \lambda \neq 1 \iff r = 1, \theta \neq 2n\pi \ (n \in \mathbb{Z}) \iff l = \theta i \ (\theta \neq 2\pi n) \iff \text{Tr } T = 2 \cosh(i\theta) \ (\theta \neq \pi k, k \in \mathbb{Z}) \iff \text{Tr } T = 2 \cos(\theta) \ (\theta \neq \pi k) \iff \text{Tr } T \in (-2, 2)$. For the orbits paths of a general elliptic see Figure 10 - (1).

4) As all the cases have exhausted, 4 follows easily. $\square$

4.4 Automorphisms of $\mathbb{H}$

We know that $\text{Aut}(\mathbb{H}) = \text{PSL}(2, \mathbb{R})$. So, for any $T \in \text{Aut}(\mathbb{H})$, we have $\text{Tr } T \in \mathbb{R}$ and $\det T = 1$. The fixed points of $T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ are solutions of $z = \frac{az+b}{cz+d}$. So,

$$z = \frac{(a-d) \pm \sqrt{(\text{Tr } T)^2 - 4}}{2c} \quad \text{where } a, b, c, d \in \mathbb{R}.$$

Corollary 4.6. *By adapting Proposition 4.5, we have 3 cases to consider when $T \in \text{PSL}(2, \mathbb{R})$:*

1. T is Parabolic $\iff |Tr T| = 2, T \neq I$, exactly 1 fixed point in $\hat{\mathbb{R}}$.
2. T is Hyperbolic $\iff |Tr T| > 2$, exactly 2 fixed points in $\hat{\mathbb{R}}$.
3. T is elliptic $\iff |Tr T| < 2$, exactly 1 fixed in $\mathbb{H}$.

4.4.1 Action on curves and orbit paths of the elements of $\text{Aut}\mathbb{H}$

1) Parabolic:

We know from Section 4.1 that when $T \in PSL(2, \mathbb{R})$ with $|Tr T| = 2, T \neq I$ and has ∞ as its fixed point then T is a simple Euclidean translation of the form $z \mapsto z + b, b \in \mathbb{R} \setminus \{0\}$. Its orbit paths are parallel horizontal lines and it takes a vertical line to a shifted vertical line (Fig 11). From Section 4.2.1, we know that applying $z \mapsto \frac{1}{z-z_0}$ on Fig. 11 gives Fig. 12. So, orbit paths of general parabolics becomes the red parallel circle tangent at the fixed point. The action is to shift the black geodesics.

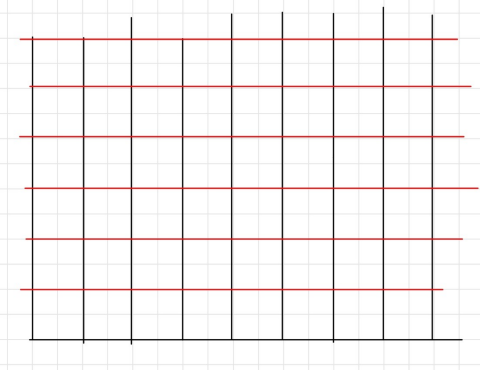


Figure 11: Orbit paths: Red parallel lines for a parabolic with fixed point ∞ . Action: take any black line to the adjacent black line.

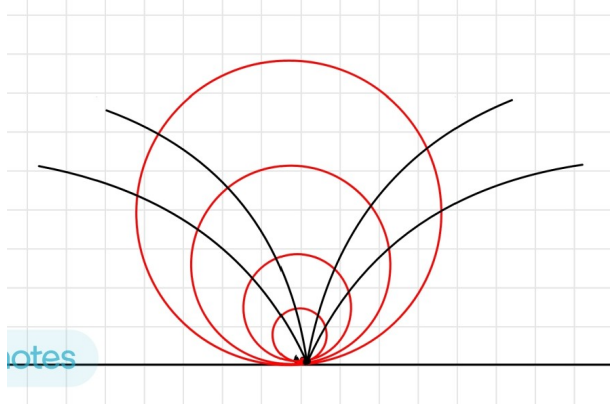


Figure 12: Orbit paths: Red circles tangent at z_0 . Action: Take the upper(lower) left geodesic to upper(lower) right geodesic.

2) Hyperbolic:

Similarly, the orbit paths for hyperbolics with fixed points 0 and ∞ are the radial lines in Fig.13 and the action is to take one black semi-circle to another. After applying $z \mapsto \frac{z-z_0^+}{z-z_0^-}$ from Section 4.2.1, we get Fig.14.

Definition 4.7. *Translation length. It is the infimum of the hyperbolic distances moved by the points after applying the hyperbolic element. Let g be a hyperbolic element. Then, the translation length T_g is*

$$T_g := \inf\{d(z, gz) : z \in \mathbb{H}^2\}$$

Definition 4.8. *Axis. We define its axis α_g as*

$$\alpha_g := \{z \in \mathbb{H} : d(z, gz) = T_g\}$$

So, take a point P on the axis. Now, we calculate $d_{\mathbb{H}}(P, T(P))$. Because $PSL(2, \mathbb{R})$ is an isometry and conjugation by an element of it leaves the hyperbolic distance unchanged, so we can assume that $T : z \mapsto \lambda z$ and $P \in i\mathbb{R}$. So, $d_{\mathbb{H}}(P, T(P)) = d_{\mathbb{H}}(it, \lambda it) = \log \lambda = l$.

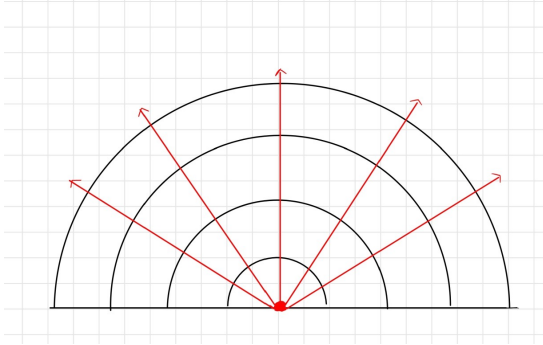


Figure 13: Orbit paths: Red radial lines for hyperbolic with fixed points $0, \infty$. Action: Take a semi-circle to an adjacent one.

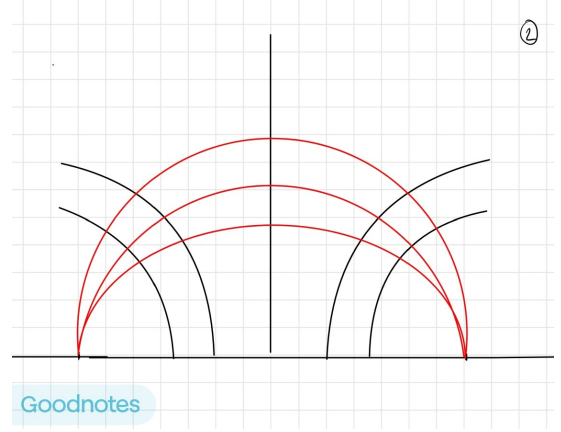


Figure 14: Orbit paths: Red semi-circles cutting x -axis at the fixed points z_0^+, z_0^- . Action: Take left/ right black geodesic to the adjacent one.

Note that this doesn't cause an issue for parabolic elements which cannot be expressed in this form as they don't have any axis.

So, what about the distance moved by the points which are not on $Ax\ T$?

Take $P \notin Ax\ T$. Now, $P = re^{i\theta}$. So,

$$d_{\mathbb{H}}(P, T(P)) = d_{\mathbb{H}}(re^{i\theta}, \lambda re^{i\theta}) = \rho \text{ where } \cosh \rho = 1 + \frac{|1 - \lambda|^2}{2\lambda \sin^2(\theta)}.$$

How do we interpret this formula?

From section 2.3, we have that $\text{Tr } T = 2 \cosh(l/2)$. Now,

$$\begin{aligned} \frac{(1 - \lambda)^2}{\lambda} &= \frac{1 + \lambda^2 - 2\lambda}{\lambda} \\ &= \frac{1}{\lambda} + \lambda - 2 \\ &= e^l + e^{-l} - 2 \\ &= 2 \cosh l - 2 \\ &= 4 \sinh^2(l/2). \end{aligned}$$

Then, using this we get that $\cosh \rho = 1 + 2 \sinh^2(\rho/2) = 1 + \frac{2 \sinh^2(l/2)}{\sin^2 \theta} \iff \sinh(\rho/2) = \sinh(l/2) \cosh(d)$. Here, d is the distance of P from the $Ax\ T$ and by *angle of parallelism* formula, we have $\cosh d = 1/\sin \theta$.

So, P shifts more away from its original position the more it was away from the axis of T and the infimum is achieved when P lies on the geodesic through the fixed points of T .

So, we get the following lemma:

Lemma 4.9. *Let $g \in PSL(2, \mathbb{R})$ be a hyperbolic element with fixed points $x_1, x_2 \in \partial \mathbb{H}$. Then, α_g is the unique geodesic passing through x_1 and x_2 and its translation length is given by*

$$T_g = 2 \cosh^{-1} \left(\frac{|Tr(g)|}{2} \right)$$

We explore the elliptic transformations in more depth now as it loses one of its fixed points when we go from $\mathbb{C}$ to $\mathbb{H}$.

4.5 Elliptic transformations

Take T to be an elliptic transformation. Then, $|Tr T| < 2$ and T has 2 fixed points in $\hat{\mathbb{C}}$ which are complex conjugates and exactly one of them is in $\mathbb{H}$. Take such a T and conjugate by an element $S \in SL(2, \mathbb{R})$ so that the fixed points are $\pm i$. Suppose, we are in this case and $T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then,

$$T(i) = i \implies \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot i = i \implies a = d, b = -c, a^2 + b^2 = 1.$$

So, we substitute $a = \cos \theta, b = \sin \theta, c = -\sin \theta$ and get $T = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$.

The action of this elliptic element on a geodesic emerging from i is to rotate it so that it fits exactly on its pair emerging from i at some angle ψ from this geodesic. We call this angle simply angle of rotation.

Now, $T'(i) = \frac{1}{(-i \sin \theta + \cos \theta)^2} = e^{2i\theta}$. So, angle of rotation ψ is 2θ here. Observing that $Tr T = 2 \cos \theta$, we get that

$$Tr T = 2 \cos \psi(T)/2.$$

Note that the angle of rotation ψ is preserved under conjugation as conjugation is a conformal transformation. We already know that the trace is preserved as well. So, this relation holds even before conjugating T with S .

4.6 Orientation reversing isometries

It is enough to produce just one orientation reversing isometry because if $\sigma, \tilde{\sigma}$ are such isometries then $\sigma \tilde{\sigma}^{-1}$ preserves orientation and we now know about every orientation preserving isometry in $\mathbb{H}$. So, we introduce the most basic orientation reversing isometry:

Consider $\sigma_0 : \mathbb{H} \rightarrow \mathbb{H}$ such that $\sigma_0(z) = -\bar{z}$. Then, σ_0 is a reflection about $i\mathbb{R}^+$, reverses orientation and is an isometry.

How do we generalize the axis which is $i\mathbb{R}^+$ here?

Consider any geodesic $\mathcal{L}$ in $\mathbb{H}$. Using Lemma 4.3, we get that the axis of $\sigma_{\mathcal{L}} = g\sigma_0 g^{-1}$ is $\mathcal{L}$ where $g \in PSL(2, \mathbb{R})$ is such that $g(i\mathbb{R}^+) = \mathcal{L}$.

Geometrically what $\sigma_{\mathcal{L}}$ does is that it reflects about the circle $\mathcal{L}$. Suppose, the center of this circle is O and has radius ρ . If P lies on a radial line emanating from O then $\sigma_{\mathcal{L}}$ sends it to P' across $\mathcal{L}$ such that $d_{\mathbb{H}}(P, \mathcal{L}) = d_{\mathbb{H}}(P', \mathcal{L})$.

We conclude this section by giving the form of any general orientation reversing isometry. It is of the following form:

$$T : z \mapsto \frac{a\bar{z} + b}{c\bar{z} + d} \quad \text{with} \quad ad - bc = 1.$$

This chapter was based on [7]. Readers can find more information about Orientation reversing isometries in it.

We finally construct our hyperbolic pair of pants. This chapter is based on [1].

5 Construction of the hyperbolic pair of pants.

5.1 Existence of a right angled hexagon with given alternate side lengths

Theorem 5.1. *Let a, b and c be in $(0, \infty)$. Then, there exists a right-angled hexagon with geodesic sides whose consecutive non-adjacent side lengths are a, b and c . Moreover, it is unique up to isometry.[1]*

Proof. Let $\mathbb{I} = \{iy : y > 0\}$ denote the imaginary axis in $\mathbb{H}$, and set $B = \{z \in \mathbb{H} : d(z, \mathbb{I}) = c\}$. By the hyperbolic distance formula, B is a one-parameter family of points and hence a one-dimensional set.

Claim 1: B is a Euclidean straight line.

For $\lambda > 0$, the dilation $z \mapsto \lambda z$ preserves $\mathbb{I}$ and hence preserves B . The only one-dimensional sets invariant under all such dilations are Euclidean lines through the origin; indeed, $d(x + iy, \mathbb{I}) = \operatorname{arcsinh}(|x|/y)$, so $d(\cdot, \mathbb{I}) = c$ is equivalent to $|x| = y \sinh c$, a pair of straight lines through the origin. This proves Claim 1.

Construction

Now, through the point i on $\mathbb{I}$, take a geodesic arc orthogonal to $\mathbb{I}$. This geodesic segment is unique by Theorem 2.8. We take this segment to be of length a .

Now, draw the unique geodesic arc γ orthogonal to the previous segment from this end point. See Figure 15 for reference.

Also, take any point p on the line B and consider the geodesic α tangent to B at p . Now, slide p on B so that

$$d(\alpha, \gamma) = \inf\{d(z, w) : z \in \gamma, w \in \alpha\} = b.$$

Shortest distance between two geodesics is realized by a common perpendicular geodesic by Section 3.7. So, b is realized by the common perpendicular from α to γ which exists and is unique by Theorem 2.9. This determines the other end point of γ and taking the unique orthogonal geodesic to $\mathbb{I}$ from p gives the remaining geodesic arc of length c of the right-angled geodesic hexagon.

Uniqueness

Take any other geodesic hexagon H' with consecutive non-adjacent sides of lengths of a, b and c . Call its sides $\gamma_1, \gamma_a, \gamma_2, \gamma_b, \gamma_3, \gamma_c$ which are ordered in left to right fashion. Apply an orientation preserving isometry m so that the segment of length a , γ_a , starts at i and is orthogonal to the $\mathbb{I}$. This is unique.

Now, the γ_2 gets mapped to the geodesic arc of which γ is a part. *Why?* This is because m is orientation preserving, conformal, takes geodesic segments to geodesic segments, preserves the length a and by Theorem 2.8. We call this β .

Now, γ_c in H' is orthogonal to γ_1 . The isometry m maps γ_1 to a segment on $\mathbb{I}$. So, m maps γ_c to a geodesic arc with one of its endpoints on $\mathbb{I}$. The other endpoint must lie on B and cut it orthogonally. Note that this hasn't fixed the segment of length c yet as there are infinitely many possibilities.

Choose any of the possibilities and set p to be the other endpoint lying on the line B . In H' , the geodesic segment γ_3 is orthogonal to γ_c and so by conformal property of m , the mapping of this segment α' must be again tangent to B . Now, α' and β have a unique common perpendicular

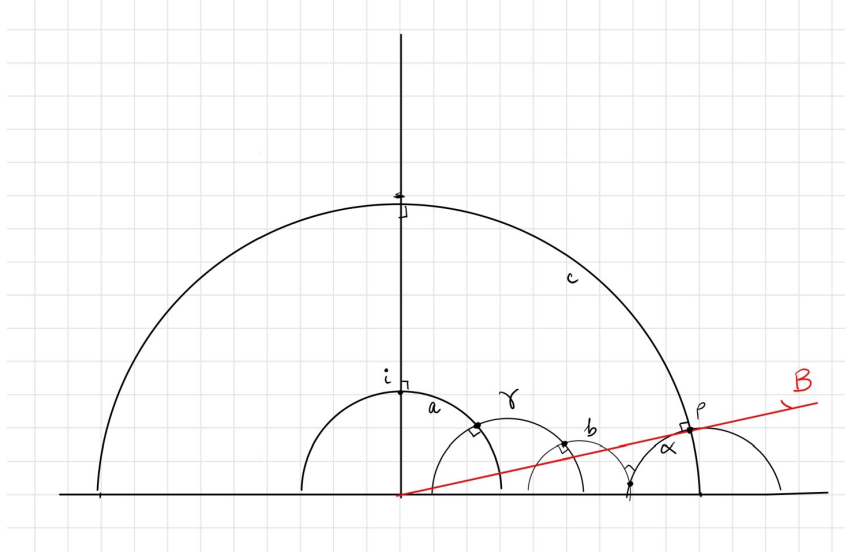


Figure 15: Right-angled geodesic hexagon constructed as above.

by Thm Theorem 2.9 and so fixing p determines both end-points of this perpendicular of some length x .

The correspondence $x \mapsto p$ is one-to-one; hence imposing $x = b$ selects a unique p and fixes γ_c . Therefore $m(H') = H$, completing the proof. $\square$

5.2 Construction of the hyperbolic pair of pants

Let G be the right-angled geodesic hexagon with consecutive sides $\alpha_1, c_3, \alpha_2, c_1, \alpha_3, c_2$ and G' be a copy of G with consecutive sides $\alpha'_1, c'_3, \alpha'_2, c'_1, \alpha'_3, c'_2$.

Now, parametrize all the sides on $[0, 1]$ with constant speeds i.e. $t \mapsto \alpha_i(t), t \mapsto \alpha'_i(t), t \mapsto c_i(t)$ and $t \mapsto c'_i(t)$ where $t \in [0, 1], i \in \{1, 2, 3\}$ have constant speeds.

Our pasting condition is as follows:

$$\alpha_i(t) = \alpha'_i(t) := a_i(t) \quad \forall i \in \{1, 2, 3\}, \quad t \in [0, 1] \quad (*)$$

Claim: $F = G + G' \bmod (*)$ is a hyperbolic surface.

The hyperbolic hexagons G and G' are hyperbolic surfaces. To rigorously get this, we use Section 3.4 and take the trivial equivalence relation that consists of only one point sets as equivalence classes i.e. $[x] = \{x\}$.

Now, we implement Theorem 3.12. Take any point on any side of G that becomes an interior point after gluing. We notice that the sum of the angles in the vertex cycle is then 2π .

For any point p that becomes a boundary point after gluing, either p is a vertex point in G or it is an ordinary boundary point. Suppose, the first case. Then, the angle made by the adjacent sides of p is $\pi/2$ as we have a right angled hexagon. See Figure 16 for reference. We glue p to $p' \in G'$ where also the angle is $\pi/2$. So, after gluing, they make an angle of π . If p is an ordinary boundary point then it doesn't glue with any point from G' in this case and so angle is $\leq \pi$ still. So, $\exists!$ hyperbolic structure such that when F equipped with this, the projection map $\sigma : G \cup G' \rightarrow F$ is a local isometry. We equip F with this hyperbolic structure.

The sides we are gluing are: $\alpha_3, \alpha'_3, \alpha_1, \alpha'_1, \alpha_2, \alpha'_2$ and any two sides lying on the same hexagon have positive distance. Moreover, F is connected. So, by Theorem 3.12, F is a hyperbolic surface. This proves our claim. $\square$

Consider the boundary curves:

$$t \mapsto \gamma_i(t) = \begin{cases} c_i(2t), & 0 \leq t \leq \frac{1}{2} \\ c_i(2-2t), & \frac{1}{2} \leq t \leq 1. \end{cases}$$

for $i \in \{1, 2, 3\}$. They form closed geodesics because they are geodesic arcs and the two points where they meet, the angle is π . If γ_1, γ_2 and γ_3 are of lengths a, b and c then we call F the *hyperbolic pair of pants* $P_{(a,b,c)}$.

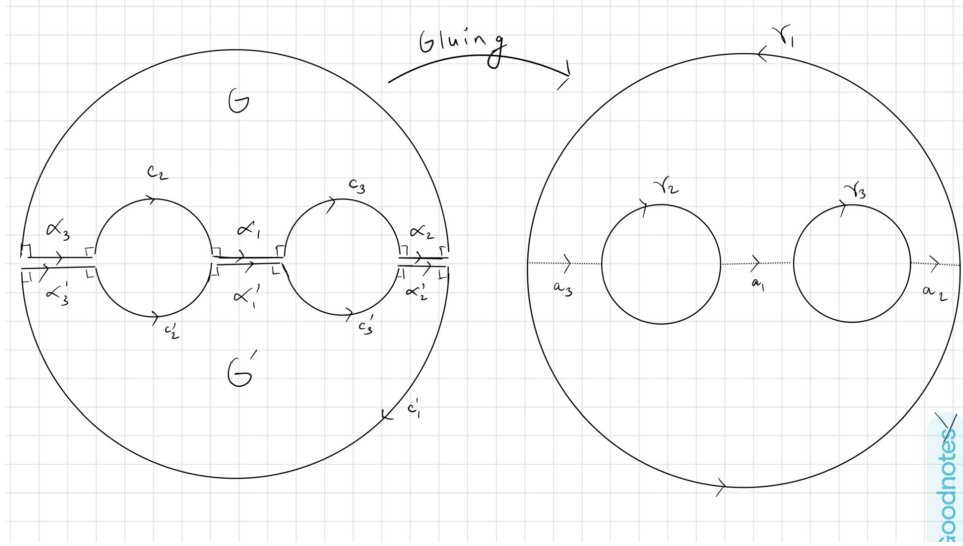


Figure 16: Construction of a hyperbolic pair of pants by gluing two right-angled hexagons along three geodesic arcs.

Proposition 5.2. *Let Y be a hyperbolic pair of pants. Then, for every pair of closed geodesic boundaries, $\exists$ a unique simple perpendicular. Cutting along these 3 perpendiculars gives us two isometric hexagons. [1]*

Claim 1: There exist three unique simple perpendiculars a_1, a_2 and a_3 between boundary geodesics in the gliding homotopy classes of simple paths between boundary geodesics.

Sketch proof of claim 1. Take a simple curve a from γ_1 to γ_2 . Now, by Theorem 3.18, we have a unique simple length minimizing perpendicular between γ_1 and γ_2 in the gliding homotopy class of a . We do a similar thing for other boundary geodesics. Hence, we have three unique simple perpendiculars a_1, a_2 and a_3 . Again, we can use second figure of Figure 16. $\square$

Claim 2: The perpendiculars are pairwise disjoint.

Proof of claim 2. Cut Y along a_3 . Notice that after cutting along a_3 , the side γ_2 is open and makes $\pi/2$ with its adjacent sides. Invoking Theorem 3.18 (B here is γ_2 and A is the closed boundary geodesic γ_3), we get that there is a length minimizing curve from the cut γ_2 to γ_3 which is also the simple, unique perpendicular between them which lies in $\text{int } Y$ except at its endpoints. Call it a'_1 . After gluing along a_3 , a'_1 is still the length minimizing path from γ_2 to γ_3 . So, it is the simple, unique perpendicular. By uniqueness, $a_1 = a'_1$. Now, a'_1 doesn't intersect a_3 or else it would not have been a path from cut γ_2 to γ_3 . Hence, $a_1 = a'_1 \implies a_1$ doesn't intersect a_3 . Same thing holds for other pairs of orthogonals. $\square$

Claim 3: Cutting along a_1, a_2 and a_3 gives us two isometric hexagons.

Suppose, we cut along a_1, a_2 and a_3 . Then, we get two geodesic hexagons G, G' with alternate sides of lengths $|a_1|, |a_2|$ and $|a_3|$. By Theorem 5.1, G and G' are isometric. $\square$

5.3 Uniqueness of the hyperbolic pair of pants

Corollary 5.3. *The endpoints of a_1, a_2 and a_3 bisect the geodesic boundaries.*

Theorem 5.4. *For $l_1, l_2, l_3 > 0$ there exists a unique hyperbolic pair of pants with closed boundary geodesics $\gamma_1, \gamma_2, \gamma_3$ of lengths l_1, l_2, l_3 up to isometry.*

Proof. We know that it exists from the existence of geodesic hexagons with consecutive alternate sides of lengths $l_1/2, l_2/2, l_3/2$ and by construction in Section 5.2.

Let $Y, \tilde{Y}$ be such (closed boundary geodesics of lengths l_1, l_2 and l_3) pairs of pants with boundary geodesics $\gamma_i, \tilde{\gamma}_i$ for $i = 1, 2, 3$ and let a_i, a'_i be the orthogonals. Then, by Claim 3 of Proposition 5.2, the orthogonals cut Y into isometric hexagons G, G' and $\tilde{Y}$ into isometric hexagons $\tilde{G}, \tilde{G}'$. The side lengths of all of them are same by corollary 5.3 and Theorem 5.1. Hence, they are all isometric to each other.

Let m be the isometry from G to $\tilde{G}$ and m' be the isometry from G' to $\tilde{G}'$. So, m sends a_i to $\tilde{a}_i$ and m' sends a'_i to $\tilde{a}'_i$ for $i = 1, 2, 3$. Combining both these isometries m, m' , we get an isometry from Y to $\tilde{Y}$. $\square$

Consider the hyperbolic pair of pants $P_{(0,0,c)}$. We know that since $P_{(0,0,c)}$ is a hyperbolic surface so the natural complete Riemannian metric g on it has curvature -1 . Now, consider $P_{(0,0,c)}^\circ$, obtained by removing the closed geodesic boundary. This surface with the restriction of the existing Riemannian metric g is no longer complete. However, as recalled in Section 1, there exists a metric conformal to g which, when defined on P° , renders the surface complete. In this chapter, we explain why this is the case. This chapter is based on Prof. Bram Petri's notes on Teichmüller Theory.[5]

6 Riemann surfaces and conformal metrics

Definition 6.1. *Riemann surface.* A smooth surface X is called a Riemann surface if it is a one-dimensional complex manifold i.e. it is Hausdorff, second-countable and equipped with an atlas $\mathcal{A} = \{(U_i, \varphi_i)\}$ so that:

1. φ_i is a homeomorphism with $\varphi_i(U_i)$ an open subset of $\mathbb{C}$.
2. for any φ_i, φ_j

$$\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \longrightarrow \varphi_j(U_i \cap U_j)$$

is a biholomorphism when $U_i \cap U_j \neq \emptyset$.

Remark 6.2. $\mathbb{H}$ is a Riemann surface. Take $\mathcal{A} = \{(\mathbb{H}, Id)\}$. Riemann surface is also 'more' than a Riemannian surface because the transition maps here are biholomorphisms whereas in case of Riemannian surfaces, there are just smooth diffeomorphisms.

Now, we state the Uniformization theorem without proof:

6.1 Uniformization theorem

Theorem 6.3. Let X be a simply connected Riemann surface (boundaryless). Then, X is biholomorphic to exactly one of $\mathbb{H}, \mathbb{C}$ and $\hat{\mathbb{C}}$.

What about Riemann surfaces which are not simply connected? They can be seen as the quotients of these 3 spaces!

For this we first state the following proposition which is related to it.

Proposition 6.4. Let D be a domain in the Riemann sphere $\hat{\mathbb{C}}$ and suppose that $G < PSL(2, \mathbb{C}) = \text{Aut}(\hat{\mathbb{C}})$ be such that

- $g(D) = D$ for all $g \in G$.
- G acts freely on D , and
- G acts properly discontinuously on D .

Then, D/G has the structure of a Riemann surface.

Actually a stronger version of the converse holds. We again state it without proof.

Theorem 6.5 (Hopf-Riemann). If X is a Riemann surface (boundaryless) and $\tilde{X}$ is its universal cover and $\pi_1(X)$ its fundamental group then $\tilde{X}$ can be given a Riemann structure so that $X = \tilde{X}/\pi_1(X)$ and $\pi_1(X)$ acts freely and properly discontinuously on $\tilde{X}$ by biholomorphisms.

Using these results, we have the following corollary:

Corollary 6.6 (Uniformization). *Let X be a boundaryless Riemann surface. Then there exists $Z \in \{\mathbb{H}, \mathbb{C}, \hat{\mathbb{C}}\}$ and a subgroup $G < \text{Aut}(Z)$ acting freely and properly discontinuously on Z by biholomorphisms such that*

$$X \cong Z/G$$

Proof. By Theorem 6.5, $X = \tilde{X}/\pi_1(X)$ s.t. $\pi_1(X)$ acts freely and properly discontinuously on $\tilde{X}$ by biholomorphisms and $\tilde{X}$ is a universal cover. Since, $\tilde{X}$ is a simply connected Riemann surface so by Uniformization theorem, it is biholomorphic to exactly one of $\mathbb{H}, \mathbb{C}$ and $\hat{\mathbb{C}}$. Taking Z to be one of these three and $G = \pi_1(X)$, we get the corollary. $\square$

Among $\hat{\mathbb{C}}, \mathbb{C}$, and $\mathbb{H}$, the upper half-plane $\mathbb{H}$ produces the widest variety of Riemann surfaces via quotient constructions.

6.2 Quotients of the 2 simply connected Riemann surfaces

6.2.1 Quotients of $\hat{\mathbb{C}}$

Proposition 6.7. *Let X be a Riemann surface. Then, the universal cover of X is biholomorphic to $\hat{\mathbb{C}} \iff X$ is biholomorphic to $\hat{\mathbb{C}}$.*

Proof. The ‘if’ direction is immediate, so we prove only the ‘only if’ part. Every $g \in PSL(2, \mathbb{C})$ has at least one fixed point in $\hat{\mathbb{C}}$ by Lemma 4.4. However, by assumption and corollary 6.6, we have

$$X = \hat{\mathbb{C}}/G$$

where $G < PSL(2, \mathbb{C})$ acts freely on $\hat{\mathbb{C}}$. This is possible only if $G = \{e\}$. $\square$

6.2.2 Quotients of $\mathbb{C}$

Proposition 6.8. *Let X be a Riemann surface. Then, the universal cover of X is biholomorphic to $\mathbb{C} \iff X$ is biholomorphic to exactly one of $\mathbb{C}, \mathbb{C} - \{0\}$ and $\mathbb{C}/\langle \begin{pmatrix} 1 & \mu \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix} \rangle$ where $\lambda, \mu \in \mathbb{C}^*$ and λ, μ are linearly independent on $\mathbb{R}$.*

Proof. First we recall that

$$\text{Aut}(\mathbb{C}) = \{\varphi : z \mapsto az + b, a \in \mathbb{C} \setminus \{0\}, b \in \mathbb{C}\}$$

We claim that G can’t have any automorphism of the form $z \mapsto az + b, a \neq 1$. If it contains such an element, then $z = b/(1 - a)$ is a fixed point of such an element contradicting freeness. So, all elements in G have the form $z \mapsto z + b$, where $b \in \mathbb{C}$. This makes G Abelian.

Claim 1: G can contain at most two $\mathbb{R}$ -linearly independent translations.

Proof. Suppose, G contains $\{z \mapsto z + b_1, z \mapsto z + b_2, z \mapsto z + b_3\}$ where $b_1, b_2, b_3 \in \mathbb{C}$, then these elements can’t be independent over $\mathbb{Q}$.

If it were the case then the span of b_1, b_2 and b_3 over $\mathbb{Z}$ would have been dense in $\mathbb{C}$. So, we could have found a sequence $(k_i, l_i, m_i) \in \mathbb{Z}^3$ such that $k_i b_1 + l_i b_2 + m_i b_3 \rightarrow 0$. This implies that

$$\varphi_{b_1}^{k_i} \circ \varphi_{b_2}^{l_i} \circ \varphi_{b_3}^{m_i}(z) \rightarrow z$$

as $i \rightarrow \infty$.

This contradicts proper discontinuity, since given any compact set K , there would be infinitely many elements of G mapping a point of K to another point of K . $\square$

So, G can have one of the three forms:

1. $G = \{e\}$
2. $G = \langle \varphi_b \rangle$ where $b \in \mathbb{C} \setminus \{0\}$.
3. $G = \langle \varphi_{b_1}, \varphi_{b_2} \rangle$ where $b_1, b_2 \in \mathbb{C}$ are independent over $\mathbb{R}$.

Quotienting by 1. gives $\mathbb{C}$ and quotienting by 3. gives us the torus. Let us focus on 2.

It is easy to see that $\mathbb{C}/\langle \varphi_b \rangle$ is biholomorphic to $\mathbb{S}^1$ through the map $[z] \mapsto \exp(2\pi iz/b)$. Also, $\mathbb{S}^1$ is biholomorphic to $\mathbb{C} - \{0\}$.

Now, let us show the converse. The cases where $X = \mathbb{C}$ and X is a torus are simple. When $X = \mathbb{C} - \{0\}$, we know that X is biholomorphic to $\mathbb{C}/\langle z \mapsto z + 1 \rangle$ by the previous argument and this proves that the universal cover is biholomorphic to $\mathbb{C}$. $\square$

Using the corollary 6.6 and the results of Section 6.2, we conclude that if a boundaryless Riemann surface X is not $\mathbb{C}, \mathbb{C} - \{0\}$, torus or $\hat{\mathbb{C}}$, then X must be a quotient of $\mathbb{H}$ by a group $G < PSL(2, \mathbb{R})$ acting freely and properly discontinuously on it by biholomorphisms. Due to this, the metric that descends down has a constant Gaussian curvature of -1.

6.3 Conformal metrics and Riemann structures

What about Riemannian surfaces?

We state it without proof only for the case when the metric has curvature -1.

Theorem 6.9. *[Killing-Hopf special case] Let (X, g) be a Riemannian surface such that g has constant curvature of -1. Then, $\exists G < PSL(2, \mathbb{R})$ such that it acts freely and properly discontinuously on $\mathbb{H}$ and $X = \mathbb{H}/G$.*

Notice that the notion of ‘Riemannian surface’ is much weaker than that of ‘Riemann surface’ but the metric of constant curvature is equipping it with the Riemann structure. Related to this, we state the following proposition:

Proposition 6.10. *Consider an orientable smooth surface Σ of finite type without boundary. Then, the following holds for it:*

$$\left\{ \text{Riemann surface structures on } \Sigma \right\} / \sim \longleftrightarrow \left\{ \begin{array}{c} \text{Complete Riemannian metrics} \\ \text{of constant curvature } -1, 0, +1 \\ \text{on } \Sigma \end{array} \right\} / \sim$$

Here, $\longleftrightarrow$ denotes one-to-one correspondence. The equivalence relation on the left is biholomorphism and the equivalence relation on the right is isometry.

Definition 6.11. *Conformal equivalence. Two Riemannian metrics ds_1^2 and ds_2^2 on a Riemannian surface X are conformally equivalent if there exists a positive function $\rho : X \rightarrow \mathbb{R}_+$ so that*

$$ds_2^2 = \rho \cdot ds_1^2$$

Why are they important? They are important because conformal classes of Riemannian metrics on an orientable boundaryless surface Σ of finite type induces a Riemann structure on Σ . More precisely,

Proposition 6.12. *Let Σ be an orientable, boundaryless surface of finite type without boundary. Then,*

$$\left\{ \text{Riemann surface structures on } \Sigma \right\} / \sim \longleftrightarrow \left\{ \begin{array}{c} \text{Conformal classes of Riemannian} \\ \text{metrics on } \Sigma \end{array} \right\} / \sim$$

Here, the equivalence on the left side is biholomorphism and the equivalence on the right side is diffeomorphism.

Remark 6.13. *Combining Proposition 6.10 and Proposition 6.12, we get that in every conformal class of Riemannian metrics on Σ , there exists a unique complete metric of constant curvature (up to scaling if the curvature is 0).*

We mention one more theorem before we start with the Brownian loop measure that will be extremely helpful in calculating the Brownian loop measure in Section 7.3.

Theorem 6.14. *Let X be a hyperbolic surface with a totally geodesic boundary. Then, there is a one-to-one correspondence among the following:*

$$\left\{ \begin{array}{c} \text{Essential free homotopy class} \\ \text{of closed curves in } X \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{Conjugacy classes of} \\ \text{hyperbolic elements in } \Gamma \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{Oriented, unparametrised} \\ \text{closed geodesics on } \mathbb{H}/\Gamma \end{array} \right\}$$

Proof. From Theorem 3.16, we have a one-to-one correspondence between first and the third set. We now show a one-to-one correspondence between second and third sets.

We do it only for the easy case where X is a closed surface so that we don't have to worry about the boundary components and the assumption that the curves are non-puncture parallel.

Let $\pi : \mathbb{H} \rightarrow \mathbb{H}/\Gamma = X$ be the covering map. Take K — a conjugacy class of hyperbolic elements in Γ . Let $g \in K$ be any element and α_g be its axis. Then, $\pi(\alpha_g)$ is a closed geodesic in X . What g does to α_g is that it shifts any point on α_g to another point on it by a distance of T_g defined in Definition 4.7. So, its action is invariant on α_g .

Now, by Definition 4.8

$$\begin{aligned} \alpha_{hgh^{-1}} &= \{z \in \mathbb{H} : d(z, hgh^{-1}z) = T_{hgh^{-1}}\} \\ &= \{z \in \mathbb{H} : d(h^{-1}z, gh^{-1}z) = T_{hgh^{-1}}\} \\ &= h(\{y \in \mathbb{H} : d(y, gy) = T_g\}) \\ &= h(\alpha_g) \end{aligned}$$

where in the third equality, we used the fact that $T_g = T_{hgh^{-1}}$ from Definition 4.7.

So, the axis of the conjugate hgh^{-1} is exactly $h(\alpha_g)$. So,

$$\pi(\alpha_{hgh^{-1}}) = \pi(h(\alpha_g)) = \pi(\alpha_g).$$

Hence, the axes of all the elements in the conjugacy class K project down to the same closed geodesic.

In the opposite direction, consider any oriented, unparametrised closed geodesic γ on $X = \mathbb{H}/\Gamma$. Then, γ lifts to a countable union of geodesics in $\mathbb{H}$. Let $\tilde{\gamma}$ be any such lift. The group $\text{Deck}(\mathbb{H}, \pi) = \Gamma$ here and so all the other lifts are from the set

$$\{h(\tilde{\gamma}) : h \in \Gamma\}$$

Now, γ belongs to some based homotopy class in Γ . Let that be g . Let the deck transformation corresponding to g be A_g . Then,

$$A_g(y) := \sigma_g(y)$$

where σ_g is defined to be the monodromy around g (defined in appendix A.5.3). The group $\langle A_g \rangle$ must leave at least one lift invariant. The only geodesic which is left invariant by the action of this group is α_g . Moreover, $\{h(\alpha_g) : h \in \Gamma\} = \{\alpha_{hgh^{-1}} : h \in \Gamma\}$. This defines the one-to-one correspondence with a conjugacy class of hyperbolic elements. $\square$

The aim here is to introduce the Brownian loop measure on Riemannian surfaces and to see how we can calculate the Brownian loop measure of the essential free homotopy classes of loops in hyperbolic surfaces which is linked to the length of the unique length-minimizing closed geodesic. In particular, it provides the tool we will use to relate $l_h(\tilde{\gamma})$ and $l_g(\gamma)$ as described in Section 1. The chapter is based on Prof. Yilin Wang's paper [9].

7 Brownian loop measure

Prior to talking about this, we need to discuss Brownian motion on Riemannian manifolds. We do this here:

7.1 Brownian motion on Riemannian manifolds

We begin by recalling the definition of Brownian motion in Euclidean space. Let $x = (x_1, \dots, x_n) \in \mathbb{R}^n$. Brownian motion starting from x is a stochastic process $(B_t)_{t \geq 0}$ such that each coordinate process (B_t^i) is an independent one-dimensional Brownian motion starting from x_i . A one-dimensional Brownian motion (B_t^i) is characterized by the following properties:

1. $B_0^i = x$
2. almost surely, the sample paths are continuous
3. $\forall s < t$, $B_t - B_s \sim \mathcal{N}(0, t - s)$ and the increment is independent of the canonical filtration $(\mathcal{F}_s)_{0 \leq s \leq t}$ generated by the Brownian motion until time t .
4. $\text{Cov}(B_t, B_s) = t \wedge s$.

The question is how to extend this definition to a Riemannian manifold (X, g) . A naive analogue such as declaring that $\text{dist}(B_t, B_s) \sim \mathcal{N}(0, t - s)$ does not make sense in general, since (X, g) may be bounded. Instead, we use the heat equation as the defining tool.

The transition density $p(\cdot, \cdot; \cdot) : \mathbb{R}^n \times \mathbb{R}^n \times (0, \infty) \rightarrow [0, \infty)$ with respect to Lebesgue measure for $W_t = B_{2t}$ is

$$p(x, y; t) = \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{\|x - y\|^2}{4t}\right).$$

If we consider the stochastic process $(W_t) = (B_{2t})$, then its transition density is the smallest positive solution of the following initial value problem:

$$\frac{\partial p}{\partial t} = \Delta p, \quad \lim_{t \rightarrow 0} p(x, y, t) = \delta_x(y).$$

Analogously, we define the transition density p_g of (W_t) to be the smallest positive solution of the following initial value problem:

$$\frac{\partial p_g}{\partial t} = \Delta_g p_g, \quad \lim_{t \rightarrow 0} p_g(x, y; t) = \delta_x(y). \quad (3)$$

Here, Δ_g is the analogue of the Laplacian called the Laplace-Beltrami operator. *But how do we generate a Brownian motion using p_g ?*

Using eq. (3), we determine the heat kernels $(p(\cdot, \cdot, t)_{t \geq 0})$ of the associated Markov semi-groups. Starting from $x \in X$, we can specify the distribution of $(W_{t_1}, \dots, W_{t_k})$ for any $t_1 \leq t_2 \leq \dots \leq t_k$ and any k through

$$\mathbb{E}[f(W_{t_1}, \dots, W_{t_k})] = \int f(y_1, \dots, y_k) p_g(x, y_1; t_1) \cdots p_g(y_{k-1}, y_k; t_k - t_{k-1}) d\text{vol}_g(y_1) \cdots d\text{vol}_g(y_k).$$

By the Chapman-Kolmogorov equation, we refine the sequence of times $0 \leq t_1 \leq \dots \leq t_k$ and define the law of W on a countable dense set of times I . Using Kolmogorov's extension theorem, we get a continuous process (W_t) which has the same finite dimensional marginals when restricted to any such set of times. Now,

$$\Delta_g(f) = \operatorname{div}_g \operatorname{grad}_g(f) = \frac{1}{\sqrt{|g|}} \sum_{i,j=1}^n \partial_i(\sqrt{|g|} g^{ij} \partial_j f)$$

If we are on a Riemannian surface (X, g) and if in a local coordinate $g = e^{2\sigma}(dx_1^2 + dx_2^2)$, then $g^{11} = e^{-2\sigma}, g^{22} = e^{-2\sigma}$ and $g^{12} = g^{21} = 0$. So, $\Delta_g = e^{-2\sigma}(\partial_{11} + \partial_{22})$ from the above formula for Δ_g . This implies that if $\sigma \in C^\infty(\Omega, \mathbb{R})$ and $\Omega \subseteq \mathbb{R}^2$ so that g is conformal to the Euclidean metric and suppose additionally that g is a complete metric on Ω with bounded curvature, then the standard Brownian motion W run at speed 2 on $\mathbb{R}^2$ and the Brownian motion $\hat{W}$ on (Ω, g) run at speed 2, both starting from $x \in \Omega$ and defined indefinitely can be coupled by a time change as follows:

Theorem 7.1. *We define for each realisation of W and $t \geq 0$,*

$$\hat{W}_t := W_s \quad \text{where} \quad s = \inf\left\{r \geq 0 : \int_0^r e^{2\sigma(W_l)} dl \geq t\right\}$$

The process that we obtain, $(\hat{W}_t)$, by time-change has the law of a Brownian motion on (Ω, g) starting from any x if $W_0 = x$ when we push forward the standard Wiener measure through the map $W \mapsto \hat{W}$.

There are two approaches to proving this result. One may adapt the rigorous proof of conformal invariance of planar Brownian motion, or alternatively compute the transition kernel of $\hat{W}$ directly and check that it satisfies eq. (3).

7.2 Brownian loop measure on Riemannian surfaces

Let (X, g) be our Riemannian surface, $x \in X, t > 0$. We define $W_x^t(X)$ to be a sub-probability measure on Brownian paths which take values in the space $C^0([0, t], X)$ run at speed 2, starting from x and well-defined on $[0, t]$. Here, well-defined means that the paths don't hit ∂X in $[0, t]$.

$$\text{Total mass : } |W_x^t(X)| = W_x(X)(\{W \mid \text{it is well-defined on } [0, t]\}) \leq 1.$$

We denote by $W_x(X)$ the probability distribution of the Brownian motion on X starting from x . We also define the measure $W_{x \mapsto y}^t(X)$ on $C^0([0, t], X)$ by

$$W_{x \mapsto y}^t(X) := \lim_{\epsilon \rightarrow 0+} \frac{W_x^t(X) \cdot 1(W_t \in D_\epsilon(y))}{\operatorname{vol}_g(D_\epsilon(y))}$$

Note that $W_{x \mapsto y}^t(X)$ differs from the Brownian Bridge measure since, we have $\operatorname{vol}_g(D_\epsilon(y))$ instead of $\mathbb{P}(W_t \in D_\epsilon(y))$. Now,

$$W_{x \mapsto y}^t(X) = \lim_{\epsilon \rightarrow 0+} \frac{W_x^t(X) \cdot 1(W_t \in D_\epsilon(y))}{\mathbb{P}(W_t \in D_\epsilon(y))} \cdot \frac{\mathbb{P}(W_t \in D_\epsilon(y))}{\operatorname{vol}_g(D_\epsilon(y))} = \mathcal{P}_{x \mapsto y, t} \cdot p_g(x, y; t) \quad (4)$$

where $\mathcal{P}_{x \mapsto y, t}$ is the Brownian bridge measure on paths from x to y well-defined on $[0, t]$. So,

$$\text{Total mass: } |W_{x \mapsto y}^t(X)| = |\mathcal{P}_{x \mapsto y, t}| \cdot p_g(x, y; t) = p_g(x, y; t).$$

Hence, it is not a probability measure.

Using the first equality in eq. (4), we get that

$$W_x^t(X) = \int_X W_{x \rightarrow y}^t(X) \, \text{dvol}_g(y).$$

To define the Brownian loop measure, we first define the parametrized Brownian loop measure on $\{W \in C([0, t], X) : t \geq 0, W_t = W_0\}$,

$$\mu_X^* = \int_0^\infty \int_X \frac{1}{t} W_{x \rightarrow x}^t(X) \, \text{dvol}_g(x) \, dt. \quad (5)$$

Consider the equivalence relation $\sim$ on the loops in X given by $W \in C^0([0, t], X) \sim \tilde{W} \in C^0([0, s], X) \iff \exists$ a continuous, increasing function $i : [0, t]_{/0 \sim t} \rightarrow [0, s]_{/0 \sim s}$ such that $\tilde{W}_{i(r)} = W_r$.

We define μ_X to be the push-forward measure of μ_X^* induced on the equivalence classes of the loops. Each equivalence class contains loops that have the same trace and orientation on X . Thus, each equivalence class consists of loops with the same trace and orientation, differing only in the choice of root and parametrization.

Proposition 7.2. *The measure μ_X satisfies the following useful properties:*

- 1) **Restriction property.** *If $X' \subseteq X$, then $d\mu_{X'}(\eta) = 1_{\{\eta \subseteq X'\}} d\mu_X(\eta)$. Here, η is an equivalence class.*
- 2) **Conformal invariance.** *If $X_1 = (X, g)$ and $X_2 = (X, e^{2\sigma}g)$ are two conformally equivalent surfaces then $\mu_{X_1} = \mu_{X_2}$.*

Proof of 1. Let η be an equivalence class such that its trace lies in X . Now,

$$\begin{aligned} d\mu_{X'}(\eta) &= d\mu_{X'}^*(\{\xi : \xi \in \eta\}) = \int_0^\infty \int_{X'} \frac{1}{t} W_{x \rightarrow x}^t(X')(\{\xi : \xi \in \eta\}) \, \text{dvol}_g(x) \, dt \\ &= \int_0^\infty \int_{X'} \frac{1}{t} W_{x \rightarrow x}^t(X')(\{\xi : \xi \in \eta\}) 1(\eta \subseteq X') \, \text{dvol}_g(x) \, dt \\ &= \int_0^\infty \int_X \frac{1}{t} W_{x \rightarrow x}^t(X)(\{\xi : \xi \in \eta\}) 1(\eta \subseteq X') \, \text{dvol}_g(x) \, dt \\ &= d\mu_X(\eta) 1(\eta \subseteq X'). \end{aligned}$$

Sketch Proof of 2.

Claim I: The measure $W_{x \rightarrow y}(X_1) = \int_0^\infty W_{x \rightarrow y}^t(X_1) \, dt$ is conformally invariant modulo time parametrisation.

Case I: Boundary of X is non-polar.

Suppose, $x \neq y$. The total mass $|W_{x \rightarrow y}(X_1)| = \int_0^\infty p_g(x, y; t) \, dt = G_g^{X_1}(x, y)$, where G_g is the Green's function on $X_1 \times X_1$ with respect to g . Why is $G_g^{X_1}(x, y)$ well-defined here? It is because since the boundary is non-polar so the Brownian motion a.s. hits it. Hence, $p_g(x, y; t)$

decays quickly when t is very large. Hence, the integral converges. *So, very long paths from x to y have a very less probability.* We use the conformal invariance of the Green's function $G_g(x, y)$ to conclude that $G_g^{X_1}(x, y) = G_g^{X_2}(x, y)$.

Also,

$$\frac{W_{x \rightarrow y}(X_1)}{|W_{x \rightarrow y}(X_1)|} = \frac{W_{x \rightarrow y}(X_2)}{|W_{x \rightarrow y}(X_2)|}$$

modulo time parametrisation. *Why?*

It is because both the measures depend only on the Brownian paths being considered from x to y in any given event E . We know from Theorem 7.1 that the trace of the paths in conformal surfaces are the same and only the time parametrisation differs. However, we are considering these quantities modulo time parametrisation. Hence, they are equal.

Combining them, we get that $W_{x \rightarrow y}(X_1) = W_{x \rightarrow y}(X_2)$.

Case II: Boundaries of X_1 and X_2 are polar or empty.

$$W_{x \rightarrow y}(X_1) = \lim_{\epsilon \rightarrow 0+} W_{x \rightarrow y}(X_1 \setminus B(x_0, \epsilon)) = \lim_{\epsilon \rightarrow 0+} W_{x \rightarrow y}(X_2 \setminus B(x_0, \epsilon)) = W_{x \rightarrow y}(X_2)$$

Here, $B(x_0, \epsilon)$ is an open ball with radius ϵ wrt g such that $x, y \notin B(x_0, \epsilon)$ and contained in both X_1 and X_2 . The second equality here follows from the fact that $\partial B(x_0, \epsilon)$ introduces a non-polar boundary in X_1 and then we are back to **Case I**.

Case III: $x = y$.

There are two types of loops whose measures we need to control to have a well-defined Green's function, ones which are very small and the other ones which are very long.

To deal with the small loops, we consider only the loops that are not entirely contained in the small ball $B(x, \epsilon)$ around x i.e. we consider the measure $W_{x \rightarrow x}(X_1)1_{(\eta \text{ not a subset of } B(x, \epsilon))}$.

What still needs to be dealt with is the measure of very long loops. So, we consider the measure $W_{x \rightarrow x}(X_1 \setminus B(x_0, \epsilon'))1_{(\eta \text{ not a subset of } B(x, \epsilon))}$ where $B(x_0, \epsilon')$ is a ball around any point $x_0 \in X$ such that $B(x_0, \epsilon') \cap B(x, \epsilon) = \emptyset$. This introduces a non-polar boundary which a.s. the Brownian motion will eventually hit and we return to case I.

What then enables us to show Claim I for $x = y$ is that

$$W_{x \rightarrow x}(X_1) = \lim_{\epsilon, \epsilon' \rightarrow 0+} W_{x \rightarrow x}(X_1 \setminus B(x_0, \epsilon'))1_{(\eta \text{ not a subset of } B(x, \epsilon))}.$$

Hence, we have shown that $W_{x \rightarrow y}(X_1) = W_{x \rightarrow y}(X_2)$ modulo time parametrisation.

Now, consider the measure on rooted loops

$$\mu_{X_1}^* := \int_X \frac{1}{t_1(\eta)} W_{x \rightarrow x}(X_1) d \text{vol}_g(x)$$

where η is any loop rooted at x and $t_1(\eta)$ is a quarter of its quadratic variation. Now,

$$\mu_{X_1}^* = \int_X \int_0^\infty \frac{1}{t_1(\eta)} W_{x \rightarrow x}^t(X_1) dt d \text{vol}_g(x)$$

Almost surely, $[W]_t = 4t$ where $[\cdot]$ represents the quadratic variation. So, $t_1(\eta) = t$ and we get that the rooted loop measure and the parametrised loop measure are the same. Hence, again forgetting the root and the parametrisation in $\mu_{X_1}^*$, we obtain μ_{X_1} .

In fact, in [9] it is mentioned that we can replace $t_1(\eta)$ by any probability density function F on roots of η and still obtain this measure. More precisely, if $\eta \sim W_{x \rightarrow x}(X_1)$ is any rooted loop, we parametrize it using its quarter-quadratic variation. Let $\theta_r \eta$ for $r \in [0, t_1(\eta)]$ denote the same loop with same orientation and parametrisation but the root at $\eta(r)$. Now, we define F to be a bounded, measurable function on rooted loops such that $\int_0^{t_1(\eta)} F(\theta_r \eta) dr = 1$ for all rooted loops η . Then,

$$\int_X F(\eta) W_{x \rightarrow x}(X_1) d \text{vol}_g(x) = \mu_{X_1}.$$

upon forgetting the parametrisation and the root (*the proof is unknown to the author*). If we put $F(\eta) = \frac{1}{t_1(\eta)}$, we get $\mu_{X_1}^*$. We need to choose F such that we can compare μ_{X_1} and μ_{X_2} . We can do so by choosing F such that the denominator is $t_2(\eta)$ - the quadratic variation of η in X_2 so that we get something related to the rooted measure $\mu_{X_2}^*$. Now, $t_2(\eta) = \int_0^{t_1(\eta)} e^{2\sigma(\eta(t))} dt$ from Theorem 7.1. So, we take

$$F(\eta) = \frac{e^{2\sigma(\eta(0))}}{\int_0^{t_1(\eta)} e^{2\sigma(\eta(t))} dt}$$

to make it a probability distribution on $\theta_r(\eta)$ for $r \in [0, t_1(\eta)]$. Then,

$$\mu_{X_1}^* = \int_X \frac{e^{2\sigma(\eta(0))}}{\int_0^{t_1(\eta)} e^{2\sigma(\eta(t))} dt} W_{x \rightarrow x}(X_1) d \text{vol}_g(x) = \int_X \frac{1}{t_2(\eta)} W_{x \rightarrow x}(X_2) d \text{vol}_{e^{2\sigma}g}(x) = \mu_{X_2}^*$$

Hence, $\mu_{X_1} = \mu_{X_2}$. □

Proposition 7.3. *Consider a Riemann surface X with non-empty boundary. Let $X' = X \setminus \partial X$ be the Riemann surface without its boundary. Then, $\mu_X = \mu_{X'}$.*

Remark 7.4. *Equipping a Riemann surface with its unique metric of constant curvature g (Remark 6.13) has an advantage: the conformal invariance of Brownian loop measure ensures that it doesn't change when we equip with this metric and calculating the Laplace Beltrami operator Δ_g is much simpler here and so p_g has a much simpler form.*

7.3 Brownian loop measure of a non-trivial essential free homotopy class of closed curves on hyperbolic surfaces

Suppose, X is a Riemann surface without boundary endowed with the unique complete hyperbolic metric. From Section 6, we get that $X = \mathbb{H}/\Gamma$, where Γ is a Fuchsian group.

Consider any non-trivial essential free homotopy class of closed curves in X . Then, by Theorem 3.16, $\exists!$ closed geodesic γ in the essential free homotopy class. Let this class be denoted by $C_X(\gamma)$. We are in the case where winding number is 1.

Theorem 7.5. *For all $x \in X, t > 0$,*

$$W_{x \rightarrow x}^t(X)(C_X(\gamma)) = \sum_{h \in [\gamma]} |W_{z \rightarrow h \cdot z}^t(\mathbb{H})|$$

where $[\gamma]$ is the conjugacy class of γ in Γ and $z \in \mathbb{H}$ is any lift of x .

Sketch Proof.

Step I: Lifting of Brownian motion. Brownian motion depends only on the local geometry of the surface X and we know from the theory on Riemann covering maps that the elementary neighborhood around x lifts to a countable number of elementary sheets each of which is homeomorphic and isometric to the elementary neighborhood. Hence, the Brownian path in the elementary neighborhood lifts to a Brownian path in the elementary sheet and the entire Brownian motion from x lifts to a unique Brownian motion in $\mathbb{H}$ from z (similar to the proof of path lifting but out of my scope).

So, the measure on Brownian paths $W_x^t(X)$ lifts to $W_z^t(\mathbb{H})$. To form a closed loop downstairs which is a member of the class $C_X(\gamma)$ and passes through x , the end point needs to be of the form $h \cdot z$ where $h \in [\gamma]$ (conjugacy class of γ).

Step II: Lifting of essential free homotopy class. Using homotopy lifting, the entire essential free homotopy class of closed loops rooted at x lifts to the sets of curves from z to $h \cdot z$ where $h \in [\gamma]$. So, we get after disintegration with respect to the possible endpoints:

$$\begin{aligned} W_{x \rightarrow x}^t(X)(C_X(\gamma)) &= \lim_{\epsilon \rightarrow 0+} W_x^t(X)(W \in C^0([0, t], X) | W_t \in D_{\epsilon, X}(x), \exists \eta \in C_X(\gamma), d_X^{loop}(W, \eta) < \epsilon) \\ &= \lim_{\epsilon \rightarrow 0+} \sum_{h \in [\gamma]} W_z^t(\mathbb{H})(\{W \in C^0([0, t], \mathbb{H}) | W_t \in D_{\epsilon, \mathbb{H}}(h \cdot z)\}) / \text{vol}(D_{\epsilon, \mathbb{H}}(h \cdot z)) \\ &= \sum_{h \in [\gamma]} |W_{z \rightarrow h \cdot z}^t(\mathbb{H})| \end{aligned}$$

Here, d_X^{loop} is the uniform sup-norm on distance in X modulo increasing reparametrization. In the third equality, we used the fact that $\mathbb{H}$ is simply connected and so all the paths from z to $h \cdot z$ are homotopic to each other and the covering map is s.t. the elementary neighborhood is isometric to the elementary sheet. The last equality follows from the computation after eq. (4). $\square$

We now quote a lemma from [9] without proof:

Lemma 7.6. *Let $X = \mathbb{H}/\Gamma$ be a complete hyperbolic surface without boundary. Let γ be an oriented closed geodesic on X of length $l(\gamma)$ and iteration number $m(\gamma)$. We have*

$$\mu_X(C_X(\gamma)) = \frac{1}{m(\gamma)} \frac{1}{e^{l(\gamma)} - 1}.$$

8 Relating the length of closed geodesics: Attempt I

Consider the hyperbolic pair of pants $X = P_{(0,0,c)}$ (Figure 17) with two cusps and a closed geodesic boundary of length c . Let γ be the closed boundary geodesic of X and let

$$C_X(\gamma) = \{\eta \mid \eta \text{ is a closed curve in } X \text{ freely homotopic to } \gamma\}$$

We have the Brownian loop measure formula only for boundaryless hyperbolic surfaces from Lemma 7.6 but this is not a problem since from Proposition 7.3, the Brownian loop measure on a surface is same as that on the same surface with its boundary removed. So, $\mu_X(C_X(\gamma)) = \mu_{\hat{X}}(C_{\hat{X}}(\gamma))$. Since, $(X^\circ, g|_{X^\circ})$ is an orientable, boundaryless Riemann surface of finite type so by Remark 6.13 $\exists!$ h (hyperbolic metric) conformal to g which makes it complete. Then, by Lemma 7.6

$$\mu_{\hat{X}}(C_{\hat{X}}(\gamma)) = \frac{1}{e^{l_h(\tilde{\gamma})} - 1} \quad (6)$$

where $\tilde{\gamma}$ is the unique closed geodesic in the essential free homotopy class $C_{\hat{X}}(\gamma)$ in the surface $\hat{X}$ equipped with the metric h . The length $l_h(\tilde{\gamma})$ is computed using h . See Figure 18 for (X°, h) .

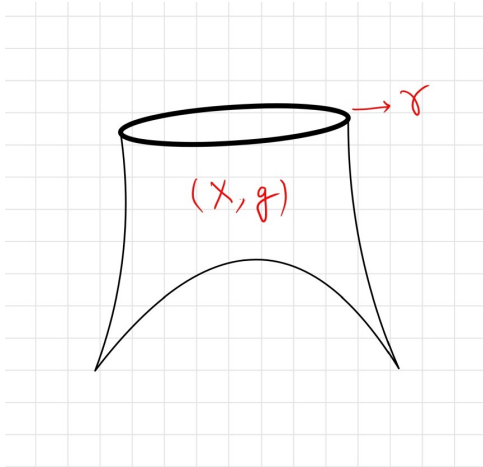


Figure 17: The hyperbolic pair of pants X with closed boundary geodesic γ of length c .

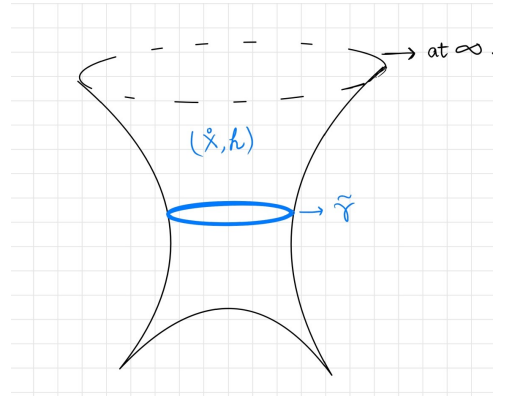


Figure 18: The interior of X equipped with h takes the open end to ∞ and make it complete.

The problem is that we want to express $l_h(\tilde{\gamma})$ in terms of $l_g(\gamma)$. Here is our first attempt.

Consider X and the hyperbolic funnel F with closed geodesic boundary γ' of length $l(\gamma)$ with opposite orientation. Now, paramterise γ, γ' on $[0, 1]$ with constant speed and define the pasting condition:

$$\gamma(t) = \gamma'(t), \quad t \in [0, 1] \quad (*)$$

and denote $Z = X + F \text{ mod } (*)$.

Claim: Z is a hyperbolic surface.

Proof. We apply Theorem 3.12 to prove this. There are no vertex points or boundary points in Z (see Figure 19). Let p be any interior point in Z and suppose that the vertex cycle $\{p_1, p_2\}$ consists of points lying on F and X . Then, p_1, p_2 must be boundary points. So, the interior angles are π and π . Hence, total interior angle is 2π . Also, Z is connected and both X and F have one side each. $\square$

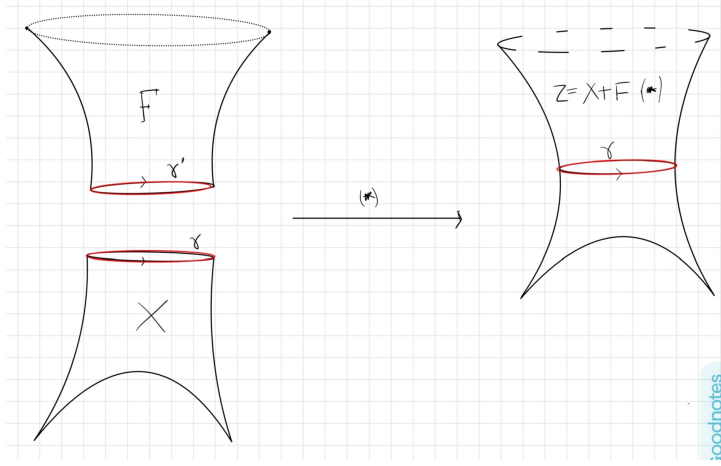


Figure 19: Illustration of the surface $Z = X + F \bmod (*)$

Now,

$$\mu_Z(C_Z(\gamma)) = \mu_Z(C_X(\gamma)) + \mu_Z(C_F(\gamma)) + \mu_Z(C_{X \cap F}(\gamma)) \quad (7)$$

Here, $\mu_Z(C_{X \cap F}(\gamma))$ represents the Brownian loop measure on Z of the closed curves lying on both X and Z and freely homotopic to γ . We let $\mu_Z(C_F(\gamma))$ represent the Brownian loop measure on Z of the closed curves restricted to F and freely homotopic to γ .

Moreover, by the restriction property of the Brownian loop measure in Proposition 7.2,

$$\mu_Z(C_X(\gamma)) = \mu_X(C_X(\gamma)) \quad \& \quad \mu_Z(C_F(\gamma)) = \mu_F(C_F(\gamma)).$$

Hence, we get that

$$\mu_Z(C_Z(\gamma)) > \mu_X(C_X(\gamma)). \quad (8)$$

Now,

$$\mu_Z(C_Z(\gamma)) = 1/(e^{l_g(\gamma)} - 1). \quad (9)$$

because the unique closed geodesic in $C_Z(\gamma)$ is still γ as it was the common closed geodesic of both the surfaces. *Why is the length of γ same after gluing?* This is because from Theorem 3.12, the projection map $\sigma : X \cup F \rightarrow Z$ is a local isometry and so with the induced metric, Z is locally isometric to $X \cup F$. In $X \cup F$, length of γ is $l_g(\gamma)$ and local isometry preserves lengths.

Using this fact, eq. (8) and eq. (9), we get that

$$\boxed{l_h(\tilde{\gamma}) > l_g(\gamma)} \quad (10)$$

The previous attempt gave only a rough estimate. We now refine the computation.

9 Relating the length of closed geodesics: Attempt II

In eq. (7), if we find $\mu_Z(C_F(\gamma))$ and incorporate this value, then we will be one step closer.

Consider the fundamental domain D of the funnel F whose closed geodesic is γ' and has length c . Here, s_1 is identified with s_3 . We shall show that, in this setting, the funnel F is biholomorphic to an annulus A_r whose Brownian loop measure is easier to calculate and then we can use the conformal invariance of the Brownian loop measure.

So, consider D in $\mathbb{H}$ but this time equipped with the flat metric!

Claim: The domain D is biholomorphic to a rectangle $\mathcal{R}$ of side lengths c and $\pi/2$, where the two sides of length $\pi/2$ are identified.

Proof. Call this rectangle $\mathcal{R}$. Take $H : D \rightarrow \mathcal{R}$ such that $H(z) = \log(z)$. We need to define it properly because $\log(z)$ has multiple values.

Take $z = re^{i\theta} \in D$. Then,

$$1 \leq r \leq e^c \quad \text{and} \quad 0 \leq \theta \leq \pi/2. \quad (11)$$

We define $\log(z) = \log(re^{i\theta}) = \log r + i\theta$. So, we stay in this particular branch when we take $\log$.

From eq. (11), $0 \leq \log(r) \leq c$ and so $H(D) = \mathcal{R}$ where the two sides of length $\pi/2$ are identified.

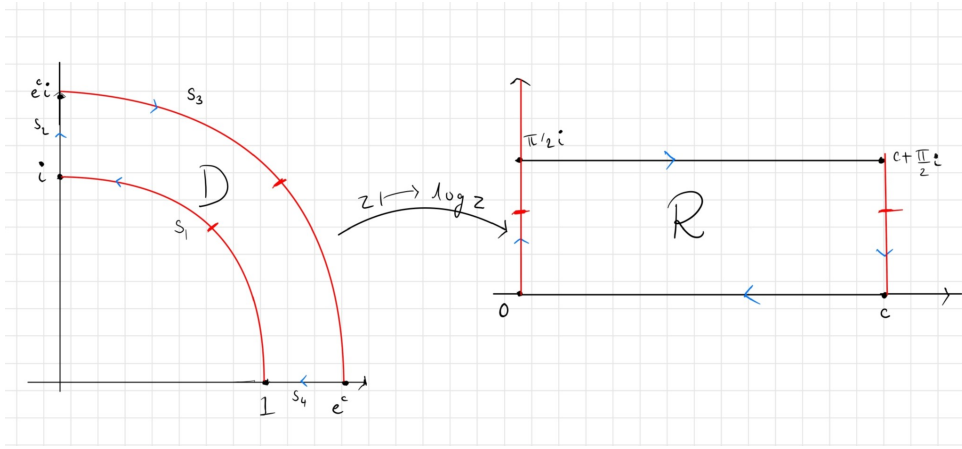


Figure 20: The fundamental domain D of the funnel F , now equipped with the flat metric, is biholomorphic to a rectangle $\mathcal{R}$ of side lengths c and $\pi/2$, with the two sides of length $\pi/2$ identified.

Claim 1: H is one-one.

Proof. Take $z_1 = r_1 e^{i\theta_1}, z_2 = r_2 e^{i\theta_2} \in D$ such that $\log(z_1) = \log(z_2)$. Then,

$$\log(r_1) + i\theta_1 = \log(r_2) + i\theta_2 \implies \log(r_1) - \log(r_2) = i(\theta_2 - \theta_1) \implies \theta_2 = \theta_1 \quad \text{and} \quad r_2 = r_1.$$

Claim 2: H is surjective.

Proof. Take $z \in \mathcal{R}$. So, $z = x + iy$ where $0 \leq x \leq c$ and $0 \leq y \leq \pi/2$. Consider e^{x+iy} . Then, $H(e^{x+iy}) = \log(e^{x+iy}) = x + iy$.

Claim 3: H is a holomorphism.

Proof. Take $z = x + iy$. Then, $H(x + iy) = \log \sqrt{x^2 + y^2} + i \tan^{-1}(y/x)$. Let $u(x, y) = \log(\sqrt{x^2 + y^2})$ and $v(x, y) = \tan^{-1}(y/x)$. Now,

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{x}{(x^2 + y^2)}, & \frac{\partial u}{\partial y} &= \frac{y}{x^2 + y^2} \\ \frac{\partial v}{\partial y} &= \frac{x}{(x^2 + y^2)}, & \frac{\partial v}{\partial x} &= -\frac{y}{(x^2 + y^2)} \end{aligned}$$

Clearly, the Cauchy-Riemann equations are satisfied: $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ and all the first order partial derivatives are continuous. So, H is a biholomorphism. $\square$

$\mathcal{R}$ is not a rectangle but a cylinder with height $\pi/2$ and circumference c . Now, consider the annulus A_r of inner radius $\exp(-r) = \exp(-\pi^2/c)$ units and outer radius 1 unit. Here, $r = \pi^2/c$. Consider the map $G : \mathcal{R} \rightarrow A_r$ such that $G(z) = \exp(\frac{2\pi}{c} iz)$.

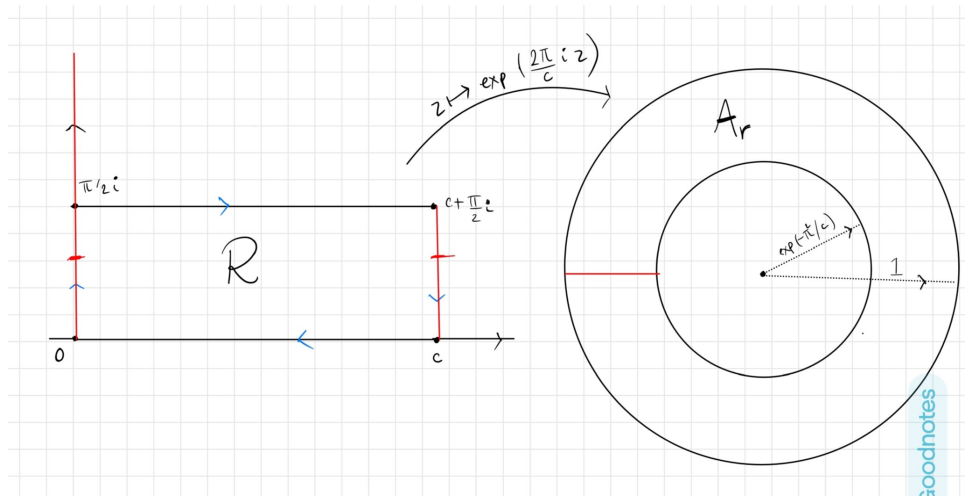


Figure 21: The rectangle R obtained from D is mapped by the biholomorphism $G(z) = \exp(iz(r/\pi))$ to the annulus A_r , thereby showing that the funnel F is biholomorphic to A_r .

Claim: G is a biholomorphism.

Proof. It is easy to see that G is bijective. We now show that it is a biholomorphism. Now, $G(z) = \exp(i\frac{2\pi}{c}z)$ and so putting $z = x + iy$, we get that $G(x + iy) = \exp(-\frac{2\pi}{c}y) \cos(\frac{2\pi x}{c}) + i \sin(\frac{2\pi x}{c}) \exp(-\frac{2\pi}{c}y)$.

Again, the Cauchy-Riemann equations are satisfied and all the first order partial derivatives are continuous and so, G is a biholomorphism. $\square$

Combining all the claims, we have proved that the hyperbolic funnel F is biholomorphic to the annulus A_r .

A biholomorphic map gives a one-to-one correspondence between free homotopy classes of closed curves in two biholomorphic surfaces. So,

$$G \circ H(C_F(\gamma')) = C_{A_r}(\tilde{\gamma})$$

where $\tilde{\gamma}$ is the unique simple geodesic in A_r . So, to find $\mu_F(C_F(\gamma'))$ we just need to calculate $\mu_{A_r}(C_{A_r}(\tilde{\gamma}))$.

Claim:

$$\mu_{A_r}(C_{A_r}(\tilde{\gamma})) = \frac{1}{e^{2\pi^2/r} - 1}.$$

Proof. Let $D' = \{re^{i\theta} \in \mathbb{H} \mid 1 \leq r \leq e^{2\pi^2/r}, 0 \leq \theta \leq \pi\} / \sim$ where the equivalence relation identifies the two semi-circular arcs via the hyperbolic $h : z \mapsto e^{2\pi^2/r}z$ and $\mathcal{R}' = \{z \in \mathbb{H} \mid \Re(z) \in [0, 2\pi^2/r], \Im(z) \in [0, \pi]\}$. Using very similar maps to H and G , $\tilde{H} : D' \rightarrow \mathcal{R}'$ such that $\tilde{H}(z) = \log(z)$ and $\tilde{G} : \mathcal{R}' \rightarrow A_r$ such that $\tilde{G}(z) = \exp(iz(r/\pi))$, we have that A_r is biholomorphic to the hyperbolic cylinder $D' / \sim$.

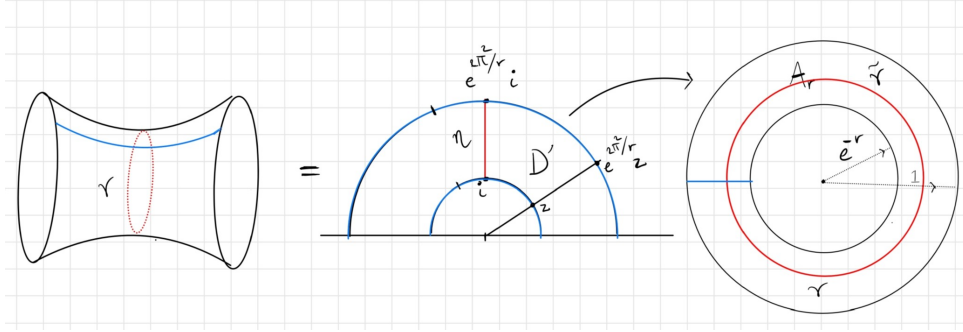


Figure 22:

The segment of the axes of the hyperbolics in a conjugacy class project down via a local isometry to a common oriented, unparametrised closed geodesic by Theorem 6.14. Using that, we get that the length of the unique simple closed geodesic γ in the hyperbolic cylinder $D' / \sim$ is $l = 2\pi^2/r$ as η projects down to γ .

So using conformal invariance of the Brownian loop measure,

$$\mu_{D'/\sim}(C_{D'/\sim}(\gamma)) = \mu_{A_r}(C_{A_r}(\tilde{\gamma})) = \frac{1}{e^{2\pi^2/r} - 1}.$$

□

Using this and putting $r = \pi^2/c$, we get that

$$\mu_F(C_F(\gamma')) = \frac{1}{e^{2c} - 1}.$$

Substituting this value in eq. (7), we get that

$$\boxed{l_h(\tilde{\gamma}) > \log(1 + e^c - e^{-c}).} \tag{12}$$

10 Another way to calculate Brownian loop measure of the funnel F

Instead of proving Lemma 7.6, we introduce a more general method to find the Brownian loop measure of funnel F based on the proof of Lemma 7.6 in [9]. The advantage is that it can be applied to settings where conformal maps may not be easy to find.

Claim: $\mu_F(C_F(\gamma')) = \frac{1}{e^{2c}-1}$.

Proof. We want to show that

$$\int_0^\infty \frac{dt}{t} \int_X W_{x \rightarrow x}^t(C_X(\gamma')) d\text{vol}_g(x) = \int_0^\infty \frac{1}{t} \sum_{h \in [\gamma]} \int_D p'_{\mathbb{H}}(z, h \cdot z; t) d\rho(z) dt = \frac{1}{e^{2c}-1}.$$

where D is a fundamental domain of the funnel F , $p'_{\mathbb{H}}$ is the transition density of the Brownian motion which gets killed as soon as it hits the lift of the geodesic boundary γ . So, in upstairs, the Brownian motion gets killed as soon as it hits $\mathbb{I}$ axis. The first equality follows from Theorem 7.5. Here, $\Gamma = \langle h_0 = z \mapsto e^c z, \sigma_0 = z \mapsto -\bar{z} \rangle$ and is not Fuchsian. Now,

$$\begin{aligned} & \mathbb{P}_x(B_t \in dx \text{ with a bounce on the boundary } \gamma) \\ &= \mathbb{P}_{\tilde{x}}(B_t \in d(h \cdot \tilde{x}) \text{ with a bounce on the } \mathbb{I} \text{ for any } h \in [\gamma]) \\ &= \mathbb{P}_{\tilde{x}}(B_t \in d(\sigma_0 h \cdot \tilde{x}) \text{ for any } h \in [\gamma]) \end{aligned}$$

where $\tilde{x}$ is a lift of x and the second equality follows from the reflection principle of Brownian motion.

So, $p_{\mathbb{H}}(\tilde{x}, h \cdot \tilde{x}; t)$ restricted to paths that bounce on $\mathbb{I}$ axis is same as $p_{\mathbb{H}}(\tilde{x}, \sigma_0 h \cdot \tilde{x}; t)$.

Using this we get that $p'_{\mathbb{H}}(z, h \cdot z; t) = p_{\mathbb{H}}(z, h \cdot z; t) - p_{\mathbb{H}}(z, \sigma_0 h \cdot z; t)$. So,

$$\mu_F(C_F(\gamma')) = \int_0^\infty \frac{1}{t} \sum_{h \in [\gamma]} \int_D [p_{\mathbb{H}}(z, h \cdot z; t) - p_{\mathbb{H}}(z, \sigma_0 h \cdot z; t)] d\rho(z) dt$$

We recognize that $h_0 \in [\gamma]$ and the axes of h_0 and σ_0 are the same and so they commute. Hence, the conjugacy class $[\gamma]$ consists of only one element i.e. h_0 .

From Buser's Geometry and Spectra of compact Riemann surfaces Ch 1, thm 1.4.2, we have that the universal cover of the funnel F is isometric to a convex domain $\hat{F}$ in $\mathbb{H}$ with piecewise geodesic boundary.

We mention a claim which is not hard to prove but skip the proof in order to be concise.

Claim: Every point in $\hat{F}$ can be pulled back into the strip domain to a point in $F_{h_0} = \{x + iy : 1 \leq y \leq e^c, x \geq 0\}$ by $G = \langle h_0 \rangle$ action so that F_{h_0} is covered entirely.

Also, $\hat{F}/\Gamma = \hat{F}/\langle h_0 \rangle = F$. So, for any measurable function f , we have

$$\int_D f(z) d\rho(z) = \int_{F_{h_0}} f(z) d\rho(z)$$

So,

$$\mu_F(C_F(\gamma')) = \int_0^\infty \frac{1}{t} \int_{F_{h_0}} [p_{\mathbb{H}}(z, h \cdot z; t) - p_{\mathbb{H}}(z, \sigma_0 h \cdot z; t)] d\rho(z) dt$$

$$\begin{aligned}
\text{Let } I_2 &= \int_0^\infty \frac{1}{t} \int_{F_{h_0}} p_{\mathbb{H}}(z, \sigma_0 e^c \cdot z; t) d\rho(z) dt \\
&= \frac{1}{2} \frac{\sqrt{2} e^{-t/4}}{(4\pi t)^{3/2}} \int_{\mathbb{R}} \int_1^{e^c} \int_{d(z, -e^c \bar{z})}^\infty \frac{r e^{-r^2/4t}}{\sqrt{\cosh r - \cosh d(z, -e^c \bar{z})}} \frac{dr dy dx}{y^2}, \quad \text{where } z = x + iy.
\end{aligned}$$

Here, we used the fact that $p_{\mathbb{H}}(z, w; t) = \frac{\sqrt{2}}{(4\pi t)^{3/2}} e^{-t/4} \int_{d(z, w)}^\infty \frac{r e^{-r^2/4t}}{\sqrt{\cosh r - \cosh d(z, w)}} dr$ and using the hyperbolic distance formula $\cosh d(z, w) = 1 + \frac{|z-w|^2}{2\Im z \Im w}$, we get that

$$\cosh d(z, -e^c \bar{z}) = -1 + 2 \cosh^2(c/2) ((x/y)^2 + 1)$$

For $u \geq 1$, we let

$$\begin{aligned}
U(u) &:= \int_{\cosh^{-1}(u^2)}^\infty \frac{r e^{-r^2/4t}}{\sqrt{\cosh r - u^2}} dr \\
&= - \int_{r=\cosh^{-1}(u^2)}^\infty \frac{1}{\sqrt{\cosh r - u^2}} d(2t e^{-r^2/4t}) \\
&= -\frac{1}{\pi} \int_{v=u}^\infty \frac{1}{\sqrt{v^2 - u^2}} d(2\pi t e^{-(\cosh^{-1} v)^2/4t}). \tag{13}
\end{aligned}$$

Then,

$$\begin{aligned}
2 \int_{F_{h_0}} p_{\mathbb{H}}(z, \sigma_0 h z) d\rho(z) &= \frac{\sqrt{2} e^{-t/4}}{(4\pi t)^{3/2}} \int_1^{e^c} \int_{\mathbb{R}} U \left(\sqrt{-1 + 2(\cosh c/2)^2 (\frac{x^2}{y^2} + 1)} \right) \frac{dx dy}{y^2} \\
&= \frac{\sqrt{2} e^{-t/4}}{(4\pi t)^{3/2}} \int_1^{e^c} \int_{\mathbb{R}} U \left(\sqrt{-1 + 2(\cosh c/2)^2 (\theta^2 + 1)} \right) \frac{d\theta dy}{y} \\
&= \frac{2e^{-t/4}}{(4\pi t)^{3/2} (\cosh c/2)} \left(\int_1^{e^c} \frac{1}{y} dy \right) \left(\int_{\sqrt{2(\cosh c/2)^2 - 1}}^\infty \frac{U(u)u}{\sqrt{u^2 + 1 - 2(\cosh c/2)^2}} du \right) \tag{14}
\end{aligned}$$

where in the third equation, we substituted $u = \sqrt{-1 + 2(\cosh c/2)^2 (\theta^2 + 1)}$.

Now, we use the fact that Abel transformation $V(v) = 2 \int_v^\infty \frac{U(u)u}{\sqrt{u^2 - v^2}} du$ has the inverse transform

$$U(u) = -\frac{1}{\pi} \int_u^\infty \frac{1}{v^2 - u^2} dV(v).$$

From eq. (13), we have that $V(v) = 2\pi t e^{-(\cosh^{-1} v)^2/4t}$ and from eq. (14), $V(\sqrt{2(\cosh c/2)^2 - 1}) = \left(\int_{\sqrt{2(\cosh c/2)^2 - 1}}^\infty \frac{U(u)u}{\sqrt{u^2 + 1 - 2(\cosh c/2)^2}} du \right)$. So,

$$2\pi t e^{-(\cosh^{-1}(2(\cosh c/2)^2 - 1))^2/4t} = 2\pi t e^{-c^2/4t} = \left(\int_{\sqrt{2(\cosh c/2)^2 - 1}}^{\infty} \frac{U(u)u}{\sqrt{u^2 + 1 - 2(\cosh c/2)^2}} du \right).$$

Plugging this value in eq. (14), we get that

$$2 \int_{F_{h_0}} p_{\mathbb{H}}(z, \sigma_0 h z) d\rho(z) = \frac{1}{4\sqrt{\pi t}} e^{-t/4} e^{c^2/4t} \frac{c}{\cosh c/2}$$

and so,

$$\begin{aligned} & 2 \int_0^{\infty} \frac{1}{t} \int_{F_{h_0}} p_{\mathbb{H}}(z, \sigma_0 h z) d\rho(z) dt \\ &= \int_0^{\infty} \frac{1}{t} \frac{1}{4\sqrt{\pi t}} e^{-t/4} e^{-c^2/4t} \frac{c}{\cosh c/2} dt \\ &= \frac{c}{\cosh c/2} \frac{1}{4c} e^{-c/2} \left[\operatorname{Erf} \left(\frac{c-t}{2\sqrt{t}} \right) - e^c (\operatorname{Erf} \left(\frac{c+t}{2\sqrt{t}} \right)) \right]_0^{\infty} \\ &= \frac{1}{e^c + 1}. \end{aligned}$$

Here, the facts that $\operatorname{Erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$ and

$$\partial_t (\operatorname{Erf} \left(\frac{L \pm t}{2\sqrt{t}} \right)) = \frac{-L \pm t}{2\sqrt{\pi} t^{3/2}} e^{-t/4} e^{-L^2/4t} e^{\mp L/2}.$$

Hence, $I_2 = \frac{1}{2(e^c + 1)}$. Similarly, we can show that $I_1 = \frac{1}{2(e^c - 1)}$. So,

$$\mu_F(C_F(\gamma')) = I_1 - I_2 = \frac{1}{2(e^c - 1)} - \frac{1}{2(e^c + 1)} = \frac{1}{e^{2c} - 1}.$$

□

A Topological tools

Based on Prof. Johnny Evan's lectures on 'Topology and Groups'.

A.1 Homotopy

Let X be a topological space.

Definition A.1. *Path.* A path in X is a continuous map $\gamma : [0, 1] \rightarrow X$.

Definition A.2. *Loop.* A path γ is a loop if $\gamma(0) = \gamma(1)$. It is a loop based at x if $\gamma(0) = \gamma(1) = x$.

For our problem, we are interested in curves that can be transformed continuously to the closed geodesic of the hyperbolic pair of pants. This is what we call — *the free homotopy class of the closed geodesic*. We now make this notion rigorous.

For this, take two loops γ_0 and γ_1 .

Definition A.3. *Free homotopy.* γ_0 is freely homotopic to γ_1 if $\exists$ a continuous map $H : [0, 1] \times [0, 1] \rightarrow X$ such that $\gamma_s(t) = H(s, t)$ is a loop for each fixed $s \in [0, 1]$.

The fact that γ_0 can be transformed into γ_1 using continuous loops that vary continuously is captured by the continuity of H in both variables.

We need a slightly more restricted notion of homotopy because we want to use algebra to study loops and a good candidate for the group operation is 'concatenation', which is possible when the loops have a common point.

Definition A.4. *Based homotopy at $x \in X$.* There is a homotopy based at x from γ_0 to γ_1 if $\exists$ a continuous map $H : [0, 1] \times [0, 1] \rightarrow X$ such that $\gamma_s(t) = H(s, t)$ is a loop for each $s \in [0, 1]$ and $\gamma_s(0) = \gamma_s(1) = x$ for all $s \in [0, 1]$.

In this case, we say γ_0 is homotopic to γ_1 and write $\gamma_0 \simeq \gamma_1$.

Lemma A.5. *If γ is a loop in $\mathbb{R}^n$ based at 0 then it is based homotopic to the zero loop $t \mapsto 0$.*

Proof. Take $H(s, t) = (1-s)\gamma(t)$. Then, H is continuous in both s and t , $H(s, 0) = 0 = H(s, 1)$, and for each fixed s , $H(s, \cdot)$ traces out a loop. Moreover, $H(1, t) = 0$ and this proves the lemma. $\square$

A.2 Fundamental group

Now, we form the group using concatenation. Let Ω_x be the set of all loops based at x . Observe that 'based homotopy' is an equivalence relation on Ω_x . **Reflexivity:** Take $H(s, t) = \gamma(t)$ for all $s \in [0, 1]$. **Symmetry:** Suppose $\gamma_0 \simeq \gamma_1$ at $x \in X$. Take H to be the homotopy. Let $H'(s, t) = H(1-s, t)$. Then, $\gamma_1 \simeq \gamma_0$. **Transitivity:** If H_1 and H_2 are the based homotopies at x from γ_0 to γ_1 and from γ_1 to γ_2 respectively, define

$$H(r, t) = \begin{cases} H_1(2r, t), & r \leq \frac{1}{2}, \\ H_2(2r - 1, t), & r \geq \frac{1}{2}. \end{cases}$$

We define $\pi_1(X, x)$ to be $\Omega_x / \simeq$. Now, we introduce the group structure on it. We define

$$[\alpha] \cdot [\beta] := [\alpha \cdot \beta],$$

where the operation on the left is the group operation that we are defining and on the right, the operation is concatenation. It is easy to show that this is a well-defined operation and satisfies the group axioms, and that the identity element is $t \mapsto x$.

So, do we have anything common between $\pi_1(X, x)$ and $\pi_1(X, y)$ if x, y are in X ?

Yes! They are the same group if there is a path from x to y . We make this more concrete now. Let δ be a path from x to y and $F_\delta : \pi_1(X, y) \rightarrow \pi_1(X, x)$ be such that $F_\delta([\gamma]) = [\delta^{-1} \cdot \gamma \cdot \delta]$. We claim the following:

Theorem A.6. F_δ is an isomorphism.

Proof. Well-defined: Take γ_0 and γ_1 in $[\gamma]$ and suppose $H(s, t) = \gamma_s(t)$. Now, $F_\delta(\gamma_0) = \delta^{-1} \cdot \gamma_0 \cdot \delta$ and $F_\delta(\gamma_1) = \delta^{-1} \cdot \gamma_1 \cdot \delta$, and $F_\delta(\gamma_0)$ is based homotopic at x to $F_\delta(\gamma_1)$ through $\delta^{-1} \cdot \gamma_s \cdot \delta$. So, $[F_\delta(\gamma_0)] = [F_\delta(\gamma_1)]$ and well-definedness follows.

Homomorphism: Up to homotopy,

$$F_\delta(\beta \cdot \alpha) = \delta^{-1} \cdot \beta \cdot \alpha \cdot \delta \simeq (\delta^{-1} \cdot \beta \cdot \delta) \cdot (\delta^{-1} \cdot \alpha \cdot \delta) = F_\delta(\beta) \cdot F_\delta(\alpha).$$

Hence $F_\delta([\beta \cdot \alpha]) = F_\delta([\beta]) \cdot F_\delta([\alpha])$.

Invertibility: Now, $F_{\delta^{-1}}(F_\delta([\gamma])) = [\gamma]$, so $F_\delta^{-1} = F_{\delta^{-1}}$. □

Lemma A.7. If the loop γ_0 is freely homotopic to the loop γ_1 based at x via γ_s , then $[\gamma_0] = [\delta^{-1} \cdot \gamma_1 \cdot \delta]$ where δ is the path $s \mapsto \gamma_s(0)$ traced out by the free homotopy.

As we will see, the fundamental group is preserved under homotopy and so if we are given a complicated surface like the ‘Pair of Pants’, we can figure out its fundamental group easily by studying that of the *sphere with 3 holes*, as they are homotopy equivalent. We make this idea precise but for that we need to know about the homotopy of functions.

A.3 Homotopy of maps

Let X, Y be two topological spaces and consider two continuous functions $F : X \rightarrow Y$ and $G : X \rightarrow Y$.

Definition A.8. *Homotopy of functions.* F, G are homotopic if $\exists$ a continuous function $H : [0, 1] \times X \rightarrow Y$ such that $H(0, x) = F(x)$ and $H(1, x) = G(x)$ for all $x \in X$.

Observe that taking $X = \mathbb{S}^1$ gives that F, G are homotopic loops.

Definition A.9. *Homotopy rel A.* F, G are homotopic rel A if $\exists$ a continuous function $H : [0, 1] \times X \rightarrow Y$ such that $H(0, x) = F(x)$ and $H(1, x) = G(x)$ for all $x \in X$, and $H(s, x)$ does not depend on s when $x \in A$.

A.4 Homotopy Equivalence

Definition A.10. *Homotopy equivalence.* A continuous function $F : X \rightarrow Y$ is a homotopy equivalence if $\exists$ a continuous map $G : Y \rightarrow X$ such that $F \circ G \simeq Id_Y$ and $G \circ F \simeq Id_X$.

Homotopy equivalence is like a ‘weak’ homeomorphism as F, G don’t need to be inverses of each other, but their compositions should be homotopic to the corresponding identity functions; and this weak homeomorphism preserves the fundamental group.

We still abuse and write $X \simeq Y$ when they are homotopy equivalent.

We now prove the result that we used in calculating the fundamental groups.

Theorem A.11. *Let $X \simeq Y$ via the homotopy equivalences $F : X \rightarrow Y$ and $G : Y \rightarrow X$. Then, $F_* : \pi_1(X, x) \rightarrow \pi_1(Y, F(x))$ and $G_* : \pi_1(Y, y) \rightarrow \pi_1(X, G(y))$ are isomorphisms.*

Before we prove this, we state a lemma that will imply the theorem immediately.

Lemma A.12. *Let $F : X \rightarrow X$ be a continuous map homotopic to $\text{id}_X : X \rightarrow X$, then the push-forward map $F_* : \pi_1(X, x) \rightarrow \pi_1(X, F(x))$ is an isomorphism.*

Proof. Let $F_t : X \rightarrow X$ be the homotopy from $F_0 = F$ to $F_1 = \text{id}_X$. Let δ be the path $(\delta(t)) := (F_t(x))$, i.e. the path traced out by x .

Now, $F \circ \gamma$ is freely homotopic to γ via $F_s \circ \gamma$. So, by Lemma 2.2, we have that $[F \circ \gamma] = [\delta^{-1} \circ \gamma \circ \delta]$. By the definition of the pushforward map F_* , $F_*([\gamma]) = [F \circ \gamma]$. Invoking Theorem 2.1, the map $[\gamma] \mapsto [\delta^{-1} \circ \gamma \circ \delta]$ is an isomorphism. $\square$

Now we prove Theorem 2.2.

Take $F \circ G : Y \rightarrow Y \simeq \text{Id}_Y$. By Lemma 2.3, $(F \circ G)_* : \pi_1(Y, y) \rightarrow \pi_1(Y, F \circ G(y))$ is an isomorphism. Now, $(F \circ G)_* = F_* \circ G_*$ and so $F_* \circ G_*$ is a bijection, implying G_* is injective and F_* is surjective. Similarly, considering $G \circ F$, we conclude that both F_* and G_* are bijections. Since they are homomorphisms (as discussed in the proof of Lemma 2.3), both are isomorphisms. $\square$

A.5 Covering spaces

Let X, Y be topological spaces and $p : Y \rightarrow X$ be a continuous map.

Definition A.13. *Elementary neighborhood. An open set $U \subseteq X$ is elementary if $p^{-1}(U)$ is a disjoint union of open sets, each of which is mapped homeomorphically onto U by the restriction of p .*

Definition A.14. *Covering map. p is a covering map if around each $x \in X$, we have an elementary neighborhood containing it.*

The pair (Y, p) is called a covering space of X .

Definition A.15. *Elementary sheet. Suppose V is one of the pre-images of U and it is path-connected, then we call it an elementary sheet.*

A.5.1 Path-lifting

Consider the covering space (Y, p) of X . Let $\delta : [0, 1] \rightarrow X$ be a path. Let $x = \delta(0)$ and take any $y \in p^{-1}(x)$. Then, $\exists!$ a unique lifted path $\tilde{\delta} : [0, 1] \rightarrow Y$ i.e. $p(\tilde{\delta}(t)) = \delta(t)$ such that $\tilde{\delta}(0) = y$.

In fact, something more is true. If we take any lift of δ and impose the condition on the lifted path that at one point it must equal this particular lift, then we get a unique lifted path. More precisely,

Lemma A.16. *Consider the covering space $p : Y \rightarrow X$, any connected space T and a continuous map $F : T \rightarrow X$. Suppose $\tilde{F}_1, \tilde{F}_2$ are two lifts. Then, $\tilde{F}_1 = \tilde{F}_2$ if and only if there exists $t_0 \in T$ such that $\tilde{F}_1(t_0) = \tilde{F}_2(t_0)$.*

A.5.2 Homotopy lifting

Lemma A.17. *Let $p : Y \rightarrow X$ be a covering space and suppose $\gamma_0 : [0, 1] \rightarrow X$ is homotopic rel endpoints to $\gamma_1 : [0, 1] \rightarrow X$ via the homotopy γ_s . Then, given any lift $\tilde{\gamma}_0$ of γ_0 , $\exists!$ a homotopy rel endpoints $\tilde{\gamma}_s$ from $\tilde{\gamma}_0$ to $\tilde{\gamma}_1$ such that $p(\tilde{\gamma}_s) = \gamma_s$ and $\tilde{\gamma}_s(0)$ and $\tilde{\gamma}_s(1)$ do not depend on s .*

A.5.3 Monodromy

Again consider the covering space $p : Y \rightarrow X$ and a loop γ in X based at x . We define the monodromy associated with the loop γ to be the permutation from $p^{-1}(x)$ to $p^{-1}(x)$ such that

$$\sigma_\gamma(y) = \tilde{\gamma}(1),$$

where $\tilde{\gamma}$ is the unique lift of γ with $\tilde{\gamma}(0) = y$.

Lemma A.18. *If γ_1, γ_2 are in the same based homotopy class, then $\sigma_{\gamma_1} = \sigma_{\gamma_2}$.*

Suppose we have a group G that acts on a topological space X . With this, we have the associated quotient map $q : X \rightarrow X/G$. We may ask if this is a covering map. The following lemma guarantees that.

Definition A.19. *Continuous action. Consider any group G acting on X . Denote by $\rho(g) : X \rightarrow X$ the action map associated with g . Then, the action is continuous if for all g , $\rho(g) : X \rightarrow X$ is a homeomorphism, $\rho(gh) = \rho(g) \circ \rho(h)$ and $\rho(1) = id_X$.*

Lemma A.20. *Suppose G acts both continuously and properly discontinuously on X . Then, $q : X \rightarrow X/G$ is a covering map.*

Is there an easy way to find the fundamental group in the case when G acts on X ? Yes, we can.

Theorem A.21. *Suppose that X is a path-connected space and $x \in X$ is such that $\pi_1(X, x) = \{1\}$, and G acts properly discontinuously on X . Then, $\pi_1(X/G, [x]) = G$.*

We lifted a path and a homotopy — both of which are functions. Can we lift any function? Yes, but in a very specific case.

A.6 Lifting criterion

Theorem A.22. *Consider the covering space $p : Y \rightarrow X$. Let T be a path-connected, locally path-connected space and $F : T \rightarrow X$ be a continuous map. Suppose $t \in T$ and $x \in X$ are given with $F(t) = x$. Let $y \in p^{-1}(x)$. Then, there exists a unique continuous lift $\tilde{F} : T \rightarrow Y$ with $\tilde{F}(t) = y$ and $p \circ \tilde{F} = F$ if and only if*

$$F_*(\pi_1(T, t)) \subseteq p_*(\pi_1(Y, y)).$$

A.7 Covering transformations

Definition A.23. *Covering transformation. Suppose $p_1 : Y_1 \rightarrow X$ and $p_2 : Y_2 \rightarrow X$ are two covering spaces of the same space X and $F : Y_1 \rightarrow Y_2$ is a continuous map such that $p_2 \circ F = p_1$. Then, F is a covering transformation from (Y_1, p_1) to (Y_2, p_2) .*

Definition A.24. *Covering isomorphism. Suppose the covering transformation F is also a homeomorphism. Then, F is called a covering isomorphism.*

Definition A.25. *Deck transformation.* A covering isomorphism from (Y, p) to (Y, p) is called a Deck transformation.

Definition A.26. *Deck group.* The group of all deck transformations.

The condition $p_2 \circ F = p_1$ implies that if $p_1(y) = x$ where $y \in Y_1$ then $F(y) \in p_2^{-1}(x)$. Hence, F can only map a pre-image of x in Y_1 to a pre-image of x in Y_2 . If given any pre-images y_1, y_2 of x , can we always find a covering transformation F such that $F(y_1) = y_2$?

Theorem A.27. *Let (Y_1, p_1) and (Y_2, p_2) be covering spaces of X and let $x \in X$, $y_1 \in p_1^{-1}(x)$ and $y_2 \in p_2^{-1}(x)$. Then, there exists a unique covering transformation $F : Y_1 \rightarrow Y_2$ such that $F(y_1) = y_2$ if and only if*

$$(p_1)_*\pi_1(Y_1, y_1) \subseteq (p_2)_*\pi_1(Y_2, y_2).$$

This follows simply from the lifting criterion.

Lemma A.28. *There is a covering isomorphism $F : (Y_1, p_1) \rightarrow (Y_2, p_2)$ if and only if $(p_1)_*(\pi_1(Y_1, y_1)) = (p_2)_*(\pi_1(Y_2, y_2))$.*

Definition A.29. *Normal covering space.* $p : Y \rightarrow X$ is a normal covering space if $p_*(\pi_1(Y, y))$ is a normal subgroup of $\pi_1(X, p(y))$.

Lemma A.30. *A path-connected covering space $p : Y \rightarrow X$ is normal $\iff$ Deck(Y, p) acts transitively on $p^{-1}(x)$ for some $x \in X$.*

Theorem A.31. *Let $p : Y \rightarrow X$ be a path-connected covering space. Choose any $x \in X$ and let y be its pre-image in Y . Let $G = \pi_1(X, x)$, $H = p_*\pi_1(Y, y)$ and let N_H be the normalizer of H in G . Then,*

$$\text{Deck}(Y, p) \cong N_H/H.$$

Using this, we get the lemma which we will be using:

Lemma A.32. *If $p : Y \rightarrow X$ is a simply-connected covering space then $\text{Deck}(Y, p) \cong \pi_1(X, x)$.*

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